tapioca
stations & media

tapiOca Server Manual

tapiOca: Stations & Media, the server you run at home

SERVER
1.44.4

November 2, 2026

tapiOca is in beta, for testing only, and this manual describes only the release named above.

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This manual covers Server 1.44.4, as released on November 2, 2026. It replaces every earlier edition.

About this manual

This manual covers **tapiOca server 1.44.4**, released November 2, 2026. It replaces every earlier edition: only the newest one applies.

Who it's for: the server owner, who installs tapiOca and keeps it running, and **Admins**, who run it from its page. People who only watch need the tapiOca Apps Manual.

What it covers: everything a server owner and an Admin can see and do in 1.44.4, from installing to troubleshooting. It describes only what's released. tapiOca is in beta, for testing only, so keep it up to date and use the newest edition of this manual.

Conventions

- **Bold** marks what's on screen, exactly as tapiOca's page shows it: tabs, buttons, settings and choices, such as **Setup**, **Save station** or **Shuffle**.
- `Code` marks what you type or read exactly: addresses, file names, commands and settings in a file, such as `http://192.168.1.20:3310` or `TZ`.
- Arrows show a path through menus: **Settings → Live TV & DVR**.
- Examples use made-up addresses, such as `192.168.1.20` and `tv.example.com`. Use your own.
- The Settings reference lists every setting with its choices and default.
- A dashed box marks where a screenshot will go in a later edition.

> **Tip:** A tip is good practice that saves time or trouble.

> **Note:** A note adds a detail worth knowing.

> **Important:** An important note warns of something that could lose data, expose something you meant to keep private, or stop playback.

Words used throughout

- **Station:** one of tapiOca's live, always-on streams of your own movies and shows, with a guide.
- **Media:** your movies and shows, played on demand in tapiOca's apps.
- **Media folder:** one collection of your shows or movies, either Plex's or one tapiOca reads itself.
- **The page:** tapiOca's own web page, where Admins run the server.
- **The apps:** tapiOca's own apps for Android, iPhone and iPad, and Roku.

Welcome to tapiOca

tapiOca is a media server you run at home and own. It turns your own movies and shows into live, always-on Stations with a real guide, and plays them on demand, as Media, in tapiOca's apps. There's no subscription, no cloud account and no ads. The server is free and open source, and runs in Docker on your own hardware.

What it does

- **Your own live Stations.** Pick shows and movies, and a Station plays them around the clock, in episode order or shuffled, with a guide two days ahead. Make Stations one at a time, from a filter that keeps itself up to date, from Plex collections, or dozens at once.
- **Station extras.** Marathons, movie nights and time-of-day blocks; commercials and trailers from your own folders; Station ID cards, an Intro Bumper and an Up Next Banner; a logo, name or clock in the corner; nearly 1,000 original logos; and even loudness.
- **A bad file never stops the show.** A program that fails is tried again, then a stand-in fills its time. tapiOca checks files before they air and lists what's wrong. It can ask Sonarr or Radarr to replace a file.
- **Media in tapiOca's apps.** People browse, search and resume your shows and movies, at the best picture and sound each device can play.
- **With or without Plex.** tapiOca reads your folders itself, with details from NFO files, tags and names. Use it alone, beside Plex, or for what isn't in Plex.
- **Family limits in tapiOca's apps.** Everyone signs in as themselves. Viewing Levels set what each person can see. A shared TV opens on **Who's tuning in?**, with a PIN for each person.
- **Away from home,** through a VPN or your own reverse proxy with HTTPS.
- **Works with Plex and Jellyfin.** Your Stations also show up in other apps at home, as a tuner and a guide.

tapiOca and Plex

Plex is optional. With your own Plex Media Server, tapiOca takes your shows and movies from it: details, artwork, and intro and credits markers. It reads the files straight from disk. Your Stations can also appear in Plex's **Live TV** guide, with tapiOca as a network tuner, which needs Plex Pass.

> **Note:** Plex's terms license Plex Media Server for personal use, on hardware and storage you own. Use tapiOca with your Plex server as those terms allow. Product names are the property of their owners.

Private by design

- **Yours, at home.** No cloud account, and nothing to sign up for. People sign in to your own server.

- **Nothing sent to the developer.** tapiOca connects only to addresses an Admin typed in and, if an Admin says yes, to GitHub, to see whether a new version is out.
- **What Admins see.** Admins can see what each person watches, and their stats, to fix problems and choose what to add. The privacy notice says so.
- **Open, and never locked.** The server is free, unlimited and open source.

What you need

- **Your own movies and shows,** in folders tapiOca reads itself, in your own Plex Media Server, or both. Watching your Stations in Plex needs Plex Pass.
- **A computer that runs Docker** on your home network: a NAS (TrueNAS SCALE, Synology, Unraid), a Linux server or virtual machine, a Raspberry Pi 4 or 5 with a 64-bit OS, or a Windows or Mac computer with Docker Desktop. x86-64 and ARM64 both work.
- **Access to your media folder** from that computer.
- **Enough processing power.** Each Station being watched is converted in real time. A modern 4-core processor can usually run several 720p Stations at once. A GPU (Intel Quick Sync, AMD or NVIDIA) handles many more. A Raspberry Pi or a Mac can't use its GPU, so plan on one or two Stations at smaller sizes there.

> **Tip:** Start with one Station. Once it plays well, run **Test this server** on the **Setup** tab to see how many your hardware can play at once.

Installing tapiOca

tapiOca comes as a ready-made Docker image, `ghcr.io/egadgetboy/tapioca-server`, for x86-64 and ARM64. Each version's image is the exact one that passed its tests. You need one file that tells Docker how to run it:

File	For
`tapioca-server.yaml`	TrueNAS SCALE
`docker-compose.yml`	Docker Compose, Synology, Docker Desktop and everything else
`unraid/stationplay.xml`	Unraid's **Docker** tab (in the repository's `unraid` folder)

Download the first two from the latest release on GitHub (`github.com/egadgetboy/tapioca-server/releases/latest`), under **Assets**. Both ask for `:latest`, the newest release. Docker downloads it when you install, and again only when you update, so tapiOca never needs the internet to start. Each file sets `TAPIOCA_INSTALL`, which tells tapiOca's page how your kind of install updates.

A few names stay `stationplay`, tapiOca's earlier name: the container, the app's name on TrueNAS and Synology, the Unraid template, the data folder in the examples, and the database, `stationplay.db`. Use them as shown.

> **Note:** Each system's steps follow that system's own documented way, and not every one has been confirmed on it yet. If a step doesn't match what you see, open an issue on GitHub.

Before you start

1. **Where your media lives** on the Docker host, such as `/mnt/tank/media`.
2. **The user tapiOca runs as,** as `UID:GID`, such as `1000:1000`. It must be able to read your media and write to tapiOca's data folder. `ls -ln /path/to/media` shows the owner's numbers in the third and fourth columns.
3. **With Plex: its address,** such as `http://192.168.1.10:32400`. Use the server's network IP, never `localhost`.
4. **With Plex: your Plex token.** In Plex Web, open any movie or episode and choose **··· → Get Info → View XML**. The new page's address ends with `X-Plex-Token=…`: that value is your token.

No Plex? Delete the `PLEX_URL` and `PLEX_TOKEN` lines (on Unraid, clear both), then add a Media folder on tapiOca's page after it starts.

> **Important:** Keep your Plex token private: it gives full access to your Plex server. tapiOca never shows it on its page or in its logs.

TrueNAS SCALE

This needs TrueNAS SCALE 24.10 (Electric Eel) or later.

1. **Make the data folder.** In **System → Shell**, run (with your pool's name for `tank`; paths are case-sensitive):

`sudo mkdir -p /mnt/tank/apps/stationplay/data sudo chown -R 1000:1000 /mnt/tank/apps/stationplay/data`

1. **Install the app.** Go to **Apps → Discover Apps → ⋮ → Install via YAML**. Name it `stationplay`, and paste the contents of `tapioca-server.yaml`.
2. **Change every line marked `<- CHANGE`:** the two folders (only the part before `:/data` or `:/media`), `PLEX_URL` and `PLEX_TOKEN` (or delete both), `TZ` (such as `America/Chicago`), and `user:` if your media belongs to another user. TrueNAS's apps user is `568:568`: if you use it, `chown` the data folder to it too. Leave `MEDIA_DIR` as `/media` and `TAPIOCA_INSTALL` as `truenas`.
3. **Check the GPU lines.** The Intel and AMD lines are on in this file. If the server has no Intel or AMD graphics, delete them, or the app won't start.
4. **Save.** TrueNAS downloads the image and starts tapiOca.
5. **Open** `http://<your-nas-ip>:3310`. The first-time setup starts.

Docker Compose (Linux, Proxmox, Raspberry Pi)

1. Make a folder, fetch the compose file, and make the data folder:

`mkdir -p ~/stationplay && cd ~/stationplay curl -fLO https://github.com/egadgetboy/tapioca-server/releases/latest/download/docker-compose.yml mkdir -p data && sudo chown -R 1000:1000 data # use your UID:GID`

1. Edit `docker-compose.yml`, changing every line marked `<- CHANGE`. Leave `TAPIOCA_INSTALL` as `compose`. For a GPU, uncomment its lines.
2. Run `docker compose up -d`, then open `http://<server-ip>:3310`.

- **Proxmox:** run Docker in a virtual machine or LXC container, with your media mounted inside it. For an Intel or AMD GPU in LXC, pass `/dev/dri` through first.
- **Raspberry Pi:** use a 64-bit OS. Its GPU can't be used, so start with the 480p picture size.

Synology

This needs DSM 7.2 or later, with **Container Manager**.

1. In **Control Panel → Terminal & SNMP**, turn on SSH.
2. In **File Station**, make `/volume1/docker/stationplay` with a `data` folder inside, and upload `docker-compose.yml` there.

3. Over SSH, find your IDs with `id` (such as `uid=1026 gid=100`), then run `sudo chown -R 1026:100 data` in that folder.
4. In `docker-compose.yml`, set `user: "1026:100"` and `TAPIOCA_INSTALL: "synology"`, and change the other `<- CHANGE` lines. Media is usually under `/volume1/`.
5. In **Container Manager**, open **Project → Create**, name it `stationplay`, choose that folder, and finish. (Or run `sudo docker compose up -d` there over SSH.)

tapiOca updates over SSH. On older systems without `docker compose`, use `docker-compose`, with a hyphen; to update there, run `sudo docker-compose pull`, then `sudo docker-compose up -d`. Intel models with built-in graphics can use the GPU.

Unraid

1. In the terminal, make the data folder, owned by Unraid's `99:100`, and add the template:

```
mkdir -p /mnt/user/appdata/stationplay/data chown -R 99:100 /mnt/user/appdata/stationplay/data curl -fL -o /boot/config/plugins/dockerMan/templates-user/my-stationplay.xml \ https://raw.githubusercontent.com/egadgetboy/tapioca-server/main/unraid/stationplay.xml
```

1. On the **Docker** tab, choose **Add Container**, and pick **stationplay** under **Template**.
2. Fill in your time zone, your media folder (such as `/mnt/user/media`), and Plex's address and token (or clear both). Leave `TAPIOCA_INSTALL` as `unraid`.
3. Choose **Apply**.

For an Intel GPU, turn on **Advanced View** and set **GPU** to `/dev/dri`. If the **Video encoding** line on tapiOca's **Setup** tab names a group, add `--group-add <that number>` to **Extra Parameters**. To use the compose file instead, install the **Compose Manager** plugin, follow the Docker Compose steps with `99:100`, and set `TAPIOCA_INSTALL` to `compose`.

Windows and macOS

tapiOca runs in **Docker Desktop**, and the computer must stay on and awake.

1. Install and start Docker Desktop.
2. Make a folder, such as `C:\tapioca`, with a `data` folder inside, and put `docker-compose.yml` there.
3. In it, delete the `user:` line, set `TAPIOCA_INSTALL` to `"docker-desktop"`, set your media folder with forward slashes (such as `D:/Media:/media:ro`), and change the other `<- CHANGE` lines.
4. In a terminal, in that folder, run `docker compose up -d`.

Allow port 3310 through the firewall if asked. Only an NVIDIA GPU on Windows (through WSL 2) can be used. If Plex runs on Windows, tapiOca can't read its `D:\…` paths, and streams those files from Plex instead. Docker Desktop hides each device's address, so trusted networks can't work there.

docker run

Make a folder with a `data` folder inside, then run, with your own settings:

```
docker run -d --name stationplay --restart unless-stopped --init \
  --user 1000:1000 -p 3310:3310 -p 3311:3311 \
  -e TAPIOCA_INSTALL=docker-run \
  -e PLEX_URL=http://192.168.1.10:32400 -e PLEX_TOKEN=paste-your-plex-token-here \
  -e TZ=America/Chicago -e PORT=3310 -e PUBLIC_PORT=3311 \
  -e MEDIA_DIR=/media -e HW_ACCEL=auto \
  -v "$PWD/data:/data" -v /path/to/your/media:/media:ro \
  --security-opt no-new-privileges:true --cap-drop ALL \
  --log-opt max-size=10m --log-opt max-file=3 \
  ghcr.io/egadgetboy/tapioca-server:latest
```

Save the command: you'll run it again each time you update. Without Plex, remove its two lines. Without the public port, remove `-p 3311:3311` and `-e PUBLIC_PORT=3311`. For Intel or AMD, add `--device /dev/dri:/dev/dri --group-add <group>`; for NVIDIA, `--gpus all`.

> **Important:** Keep the restart setting (`--restart unless-stopped`, or `restart: unless-stopped`). Restoring a backup restarts tapiOca, and Docker must start it again.

GPU encoding

A GPU makes tapiOca much lighter on your server, and lets more Stations play at once, 1080p included. tapiOca uses Intel and AMD graphics through VA-API (including Quick Sync) and NVIDIA through NVENC. At startup it test-encodes a short clip, and uses the first GPU that passes, NVIDIA first. Otherwise it uses the processor. The **Video encoding** line on the **Setup** tab says which, and why.

- **Intel or AMD:** the container needs `/dev/dri` and its group, under `devices:` and `group_add:`. Find the group with `ls -ln /dev/dri` (the group of `renderD128`). It's usually `107` on TrueNAS. If it's wrong, the **Video encoding** line names the right one.
- **NVIDIA:** install the NVIDIA driver and Container Toolkit (on TrueNAS, **Apps → Configure → Settings → Install NVIDIA Drivers**), then uncomment the `deploy:` block, or add `--gpus all`.
- **Choosing:** `HW_ACCEL` is `auto`, `nvidia`, `intel`, `amd` or `cpu`. With several GPUs, `HW_DEVICE` picks one.

Safeguards. A program or copy that fails on the GPU carries on from the same point on the processor. After 3 programs in a row fail on the GPU but play on the processor, tapiOca stops using it until it restarts. Stalls never count against the GPU, because a slow disk causes them too.

How tapiOca reads Plex's files

tapiOca asks Plex where each file is, then reads it straight from disk. Mount your media at `/media`, and tapiOca matches the end of each Plex path to a file there (at least the file name and its folder). The **Media files** line on the **Setup** tab shows whether it works. Where it can't see a file, it streams it from Plex, which adds load to Plex. For unusual layouts, set `PATH_MAPPINGS`, such as `"/data/media:/media"` (several pairs separated by `;`).

The two ports

- **3310** (`PORT`), the home port: the page at home, Plex's tuner, the guides, the playlist and the streams. Home network only.
- **3311** (`PUBLIC_PORT`), the public port: the page and apps from the internet, through a reverse proxy or tunnel. Remove it if you don't need it.

> **Important:** Never forward port 3310 on your router, or point a tunnel or reverse proxy at it. Its tuner and streams never ask for a password.

First-time setup

The first time someone opens tapiOca's page, the setup opens, titled **Welcome to tapiOca**. It asks a few questions and checks that everything works. Use **Next** and **Back**, or the steps down the side. Each answer is saved when you move to another step. **Skip setup** keeps everything else as it is.

> **Important:** On a new server, skipping the setup leaves no Admin. Until there is one, anyone at home can use tapiOca's page and apps and change anything. Add an Admin in step 4, or on the **Access** tab.

SCREENSHOT TO COME

The setup's first step, Checking your server, with its list of checks

The ten steps

1. **Checking your server.** tapiOca checks the server itself, never waiting on Plex: that it can read your media folder (`MEDIA_DIR`, and any folder `PATH_MAPPINGS` names) and a video in it, how it encodes video, whether its clock matches your browser's, and its backups. If nothing is at `MEDIA_DIR`, it says where your media seems to be, and what to set `MEDIA_DIR` to. Each check shows **Working**, **Needs a look**, **Not working**, **Good to know** or **Not checked yet**, and says what to do. After changing tapiOca's app settings, restart it, then choose **Check again**.
2. **How stations play.** The **Picture for new stations** and the number of **Tuners**, with **Test this server** to help you choose.
3. **New station settings.** What each new Station starts with: **Subtitles**, **Commercials & trailers**, **Station ID card**, **Intro Bumper**, the Up Next Banner, and what goes **In the corner**. Stations you already have don't change. This step is the only place to change these defaults.
4. **Who can use tapiOca.** Enter your **Name** and a **Password** to become the first Admin, and you're signed in right away. To let your home's devices skip signing in, tick **Devices on my home network skip signing in**. It offers the network tapiOca sees you on, such as `192.168.1.0/24`. Its devices then open on **Who's tuning in?** in tapiOca's apps. (A tapiOca updated with signing in off offers **Anyone on my network** or **Only people who sign in**.)

5. **Who sees what.** The Viewing Level **New people start on** (**Unrestricted** until you choose another), and each person's level.
6. **Watching away from home.** Whether tapiOca's apps can watch away from home, the address they use, and whether tapiOca can reach it.
7. **Media in tapiOca's apps.** Which Media folders the apps can browse and play (every one, to start), and **Even sound for a show's episodes**.
8. **Checking files.** When the overnight deep scan runs, or whether it runs.
9. **Tuner. Will you watch your stations in other apps?** Tick **Plex**, **Jellyfin**, **Emby**, **Kodi**, **Another app**, or **Only tapiOca's own apps** on its own. Each app ticked shows its steps and addresses, with **Copy** buttons. This only changes what the step shows: every app can still use the tuner. For Plex, three checks show whether tapiOca can reach Plex (and whether the server's owner has Plex Pass), whether it finds the files Plex lists, and whether Plex has tapiOca as a DVR yet. **Check again** runs them again.
10. **All set.** Your settings, anything that still needs a look, and **Tell me when a new tapiOca is out**. The box is never ticked for you: unticked is No.

> **Important:** Never trust a guest network in step 4. Anyone on a trusted network can use tapiOca's apps without a password.

> **Tip:** If Plex says "There was a problem fetching channel mappings," reload Plex's page and start again. Plex renames the tuner partway through, so it may still know tapiOca's tuner from before.

SCREENSHOT TO COME

The setup's Tuner step, with Plex and Jellyfin ticked and their addresses below

Running it again

Choose **Run setup again** on the **Setup** tab, or **Setup** at the bottom of the page. Your current answers are filled in. When an update brings a new question for an Admin, the setup opens once by itself, titled **What's new in tapiOca**, with just that question.

The INFO station

A new tapiOca starts with Station 99, **INFO**. Plex won't add a tuner whose guide is empty, so INFO lets you add tapiOca to Plex before you make a Station. Once you have one of your own, INFO is for tapiOca's apps only: Plex and other apps leave it out.

- **What it shows:** a film of teal pearls flying through tapiOca's ring, to a light tune, with the time and date, tapiOca's web address, and a panel that changes every 15 seconds: **What's new in tapiOca**, **tapiOca's apps, in beta**, **By the numbers** (counts of your Stations and shared Media, never titles) and **A tip for tapiOca's apps**.
- **For everyone** who can watch your Stations in the apps, whatever their Viewing Level. It never shows your server's addresses or how it's doing.
- **Always playing.** Its film comes with tapiOca, needs no Plex or internet, is never checked or listed as broken, and drifts slightly, so nothing burns in. The guide shows it as 4-hour listings.
- **Always 99.** You can rename it and change its settings, but not its number. It plays only its own film.

To remove it, delete it on the **Stations** tab. To bring it back, choose **Add the INFO station** under **The INFO station** on the **Setup** tab (another Station on 99 must move first).

tapiOca's page

tapiOca's page is where Admins run the server. Open it at `http://<server-ip>:3310` at home, or at your own away-from-home address. It fits phones, tablets and computers.

SCREENSHOT TO COME

tapiOca's page on a computer, with the header, the tabs and the Stations tab

The tabs

Tab	What it's for	Who sees it
Stations	Make, change and delete Stations; what's on; the guide.	Everyone
Setup	Tuner addresses, Media folders, picture and tuners, backups, new versions, the INFO station.	Admins
Access	People, devices, trusted networks and proxies, Viewing Levels, tuner access, away from home, the apps' limits and Media, API tokens.	Admins
Stats	How much each Station and title is watched; for Admins, the server's health and who's watching.	Everyone
Logs	What tapiOca is doing, the apps' problems, the access log, alert notifications.	Admins
Broken files	Files that can't play properly, people's reports, file checks, replacing files. Its count is what needs you.	Admins

The header has small pills: Plex's state (**Plex connected**, or what's wrong; **Nothing to play yet** without Plex or a Media folder); tuners in use ("2 of 4 tuners in use · 3 connections", amber when every tuner is busy); and, for Admins, **Away from home** (**Up**, **Down**, **Checking**, **Can't check** or **One step left**), the number of alerts, and "tapiOca 1.45.0 is out" when a new version is. Your name and role, such as "Ada · Admin", opens **Your account**.

The tab row also has a **Buy me a coffee** link, shown to everyone. It opens the developer's page on buymeacoffee.com, and is never required.

The footer shows tapiOca's name and version, **Setup**, the copyright, **GNU AGPL v3**, **Source**, **Open-source notices** and **Privacy**. **Appearance** keeps the page **Light** or **Dark** in this browser; **Automatic** follows the device.

Signing in

Once there's an Admin, everyone signs in with a **Name** and **Password**. At home, **Who's signing in?** lists the Admins who signed in on that browser with their password, each marked **PIN** (in Light sign-in) or **Password**. **Sign in with a password** is always there, for anyone. **Remove from this browser** takes you off the list. A User is never listed.

The page signs you out after an hour without activity. An Admin signed in only for watching (by a PIN away from home, or by a trusted network) sees why, with **Sign in with your password**.

Your account

Choose your name at the top of the page to **Change your password** (your **Current password** and a **New password**, which signs you out everywhere else), to choose **How you sign in** (Admins: **Full** or **Light**, with **Your password, to change this**), to **Link an app**, or to **Sign out**.

Linking an app with a code

On a TV, tapiOca's apps show a short code, such as `K7QM-4DPX`, so no one types a password with a remote.

1. On tapiOca's page, signed in with your password as the person who'll watch, choose your name, then **Link an app**. (The app may also show the page's `/link` address.)
2. Enter the **Code** and choose **Next**.
3. The page names the app, such as "tapiOca for Roku will be signed in as Ada." Link it only if it's yours and in front of you. Choose **Link**.

A code lasts 10 minutes. The app then appears under **Signed-in apps** on the **Access** tab.

The page as an app

Add tapiOca's page to a home screen, or install it on a computer, to open it in its own window: in Safari on an iPhone or iPad, **Share → Add to Home Screen**; in Chrome on Android, **Add to Home screen**; on a computer, **Install** in Chrome's or Edge's address bar, or **File → Add to Dock** in Safari. Chrome and Edge install only from an `https://` address. Nothing is stored on the device: tapiOca must be reachable.

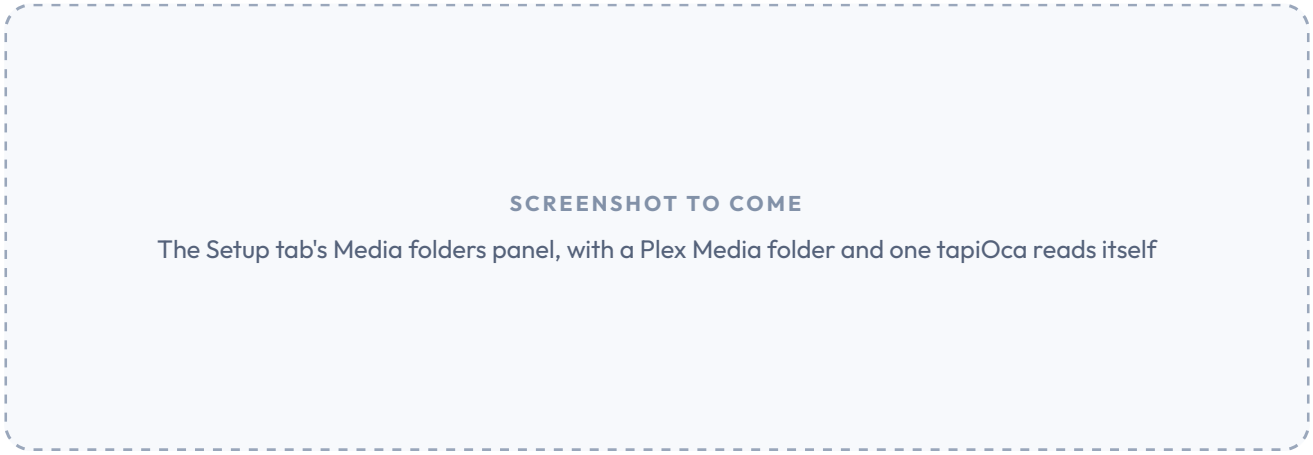
Media and Media folders

Everything tapiOca plays comes from **Media folders**, each holding shows or movies. A Media folder is either one of Plex's, or one tapiOca reads itself from folders in its media folder (`MEDIA_DIR`, `/media` in the container). Both work the same way for Stations, filters, smart Stations and Media in the apps.

The Media folders panel

On the **Setup** tab, **Media folders** lists every Media folder, Plex's and tapiOca's own.

- **The switch** keeps a Media folder out of tapiOca's apps when it's off. Stations can still use it. It saves at once.
- **Media folders tapiOca reads itself** also show their folders, when they were last read (such as "Read 5 minutes ago", or "Reading 120 of 800 files"), and **Read again**, **Edit** and **Remove**.
- **Notes** say when a folder can't be read now, or is also in one of Plex's Media folders (its titles then appear twice).



Adding a Media folder tapiOca reads itself

1. On the **Setup** tab, choose **Add a Media folder**.
2. Give it a **Name**, such as "Home Movies".
3. Under **It holds**, choose **Shows** or **Movies**.
4. Open a folder in the list, then choose **Add this folder**. Add as many as you like, but not one inside another.
5. Choose **Add**. tapiOca reads it right away.

To use tapiOca without Plex, leave out `PLEX_URL` and `PLEX_TOKEN` when you install.

Laying out its folders

Lay out folders as Sonarr, Radarr, Plex, Jellyfin and Emby do:

- **Shows:** `TV/Show (2010)/Season 01/Show (2010) - S01E02 - Title.mkv`. Also `1x02`; `S01E02E03` (two episodes in one file); `Specials` or `Season 00`; and dates for daily shows (`Show - 2024-03-15 - Title.mkv`).
- **Movies:** `Movies/Movie (1999)/Movie (1999).mkv`. Other files in a movie's folder are its versions (`Movie (1999) - 4K.mkv`). `cd1` and `cd2`, `part1` and `part2`, or `disc1` and `disc2` are parts of one movie.
- **IDs in names,** such as `{tmdb-12345}` or `{tvdb-67890}`, are kept.
- **Left out:** hidden files; extras folders (Extras, Featurettes, Behind The Scenes, Deleted Scenes, Interviews, Trailers, Shorts, Other); samples; a NAS's own folders (`@eaDir`, `#recycle`); copied discs (`BDMV`, `VIDEO_TS`); and links that lead outside the media folder.

Where its details come from

For each detail, the first of these that has it: **NFO files** beside the videos (titles, years, plots, genres, ratings, studios, cast and crew, collections); **tags inside the files**; then **folder and file names**. tapiOca looks nothing up online and never guesses. A title from a name drops its release details: `Coastline.2010.Special.1080p.WEB-DL.x264-GRP.mkv` is "Coastline 2010 Special".

- **Ratings** come from the NFO files (`<mpaa>`, such as `PG-13`). A title with no rating is unrated, so Viewing Levels that hide unrated titles hide it.
- **Pictures** come from beside the videos (`poster.jpg`, `folder.jpg`, `fanart.jpg`, `clearlogo.png`, `season01-poster.jpg`, an episode's `-thumb.jpg`), then from covers inside the files.
- **Not yet:** skipping intros and credits, and pictures for titles with none beside them or inside them.

tapiOca's apps never get a file's name or a folder's path.

Reading it again

tapiOca reads new and changed files soon after it starts, every 6 hours, when you choose **Read again**, and when Sonarr or Radarr brings a replacement it asked for. It reads at the lowest priority, and never while anyone is watching.

- **Renamed, moved or replaced files** keep their place on Stations, and what people watched of them.
- **A folder that can't be reached** keeps its programs, listed as missing until it's back. They're never taken off your Stations for it.
- **A name that isn't UTF-8,** as on some older NAS shares, is skipped, and the **Logs** tab says so. Rename it.

Media folders from Plex

With `PLEX_URL` and `PLEX_TOKEN` set, Plex's TV and movie Media folders appear by themselves, with Plex's details, artwork, ratings, and intro and credits markers. For Plex's titles, tapiOca never reads anything from file names. While Plex isn't answering, Media folders tapiOca reads itself still play.

Sharing Media with the apps

Every Media folder of shows and movies, including ones added later, is shared with the apps until an Admin saves a choice: under **Media in tapiOca's apps** on the **Access** tab, in the setup, or with the switches here. A choice is kept, even none. Viewing Levels still decide what each person sees, and Stations, Plex and other apps aren't affected.

A tapiOca updated from before 1.34.1 that never saved a choice shares nothing until an Admin chooses. The setup asks once, with every Media folder ticked.

Making Stations

There are four ways to make a Station, all on the **Stations** tab:

- **New station:** one Station, with titles you pick or a filter.
- **Smart stations:** up to 50 at once from a filter, one, or one for each decade, genre, studio and more.
- **From Plex collections:** one for each collection you pick, up to 200 at once.
- **Duplicate,** on a Station's card: a copy to change before saving.

Making a Station

1. Choose **New station**.
2. Give it a **Number** (1 to 9999) and a **Name** (up to 60 characters). With no name, it's called "Station 12", after its number.
3. Choose a **Logo** (see Station features).
4. Under **What's on this station**, choose **Pick shows & movies** or **Use a filter**.
5. Change anything under **How it plays**, **Specials**, **Between programs**, **On screen** and **When someone tunes in**. Each group shows a summary when closed.
6. Choose **Save station**.

A new Station starts with the settings chosen in the setup's **New station settings** step (see the Settings reference for each default). If you close the editor with unsaved changes, it asks first.

SCREENSHOT TO COME

The Station editor, with Pick shows & movies chosen and posters to pick from

Picking shows and movies

Choose a Media folder from the list, then choose posters to pick them. The filter box narrows the list: every word must match a title, genre or year, so "western 1959" finds Westerns from 1959. The three poster buttons change the poster size. **Everything in** a Media folder adds all of it. A title already on another Station is marked ("Already on station 12"). Below the list, tapiOca totals what you picked, in hours. The list shows up to 3,000 titles per Media folder: type to find the rest.

Using a filter

A filter asks which titles match, and keeps following the answer as your movies and shows change. Each condition narrows the results. Several values in one condition mean "any of these".

Condition	What it does
Type	**Movies** or **TV shows**.
From these Media folders	One or more.
Plex's filters	Genre, content rating, network, studio, country, collection, label, director, writer, actor and more. People are searched as you type.
Decade	For TV shows, the year they started.
Added in the last	1 to 3650 days, or **Any**. For TV, the episodes added in that time.
Minimum audience rating	**Any**, or 5 to 9 out of 10. Titles without a rating are left out.
Title contains	Words in the title.

A count of matches updates as you go. **Choose which...** lists every match, so you can uncheck any: those stay out, and new matches still join. Then choose **Add this filter**. A Station can combine up to 50 Media folders, filters and collections.

Examples: Genre: Action, for action movies. TV shows, Decade: 1950s and 1960s, for classic TV. Content Rating: TV-Y and TV-G, for kids' TV. For Christmas movies, add the label "Christmas" to them in Plex (**Edit**), then filter on Label: Christmas.

Smart stations

1. Choose **Smart stations**.
2. Set the filter.
3. Under **Make**, choose **One station**, or **One for each...** and what to split by, such as a decade or genre. tapiOca lists up to 40, with how much each has ("1960s Comedy · 88 movies"). Check the ones you want, and rename them if you like.

4. Set the **First number**. Each new Station takes the next free number from there.
5. Choose **Make station** (or **Make 10 stations**).

Each new Station gets a logo and the default settings, and keeps following its filter.

Stations from Plex collections

1. Choose **From Plex collections**.
2. Check the collections you want (**Search collections** narrows the list). One that's already a Station shows as checked.
3. Choose **Make station** (or **Make 5 stations**).

Each becomes a shuffled Station named after its collection, with the Collection logo. Titles added to or removed from the collection in Plex join or leave the Station within about an hour. If a tool makes the collection again, the Station finds it by name.

The Station card

Each Station's card shows what's on now, its description, who made it and when, and labels for anything that needs you, such as **2 broken** or **Update ready**. **Details** summarizes every setting. A red triangle means the Station can't start, is off the air, or has nothing to play.

Button	What it does
Guide	What's on now and over the next 2 days.
Edit	Opens the editor.
Duplicate	A copy with the next free number, to change before saving.
Check files	Quick-checks every program on it now.
Update now	Applies changes to what's on it at the next program break at least a minute away.
Reshuffle	Shuffled Stations only: a new random order, from the next program break.
Delete	Removes it, with **Undo** for 10 seconds. Its stats go with it.

Who can make Stations. Admins can change every Station. A User makes as many as an Admin allows (3 to start) and changes only their own. People on a Viewing Level with limits only watch. A Station can be as big as you like: one with tens of thousands of episodes takes a few seconds to build.

How Stations play

One format for each Station

Each Station is one continuous stream, as on broadcast TV. Everything on it is converted as it plays to the Station's format: H.264 at the Station's picture size, 29.97 frames a second, in standard color (SDR), with stereo sound. So a 4K movie, an old DVD copy and a commercial follow one another without a glitch, in any app.

	Size	Video per viewer	Work for the server
480p	854×480	1.5 Mbps	About half of 720p
720p (standard)	1280×720	3.5 Mbps	Light enough for most servers
1080p	1920×1080	6 Mbps	About twice 720p; a GPU helps

Sound adds 192 kbps. A new size takes effect the next time the Station starts, after everyone stops watching it.

- **4K and HDR.** Stations play at up to 1080p in standard color, on purpose: 4K HDR would need a GPU fast enough to convert everything live, and about 20 Mbps a viewer. So 4K is scaled down, and HDR (HDR10, HDR10+, HLG and most Dolby Vision) is converted to standard color the way a TV would.
- **Dolby Vision profile 5** files can't be shown correctly by anything but a Dolby Vision player, so Stations don't play them. They're listed as **Unsupported** on the **Broken files** tab.
- **Movies in several files** play back to back as one program. If one part is missing or won't play, it's listed ("part 2 of 3"), and a stand-in plays the rest of its time.

Tuners

The **Tuners** number (1 to 20, under **Picture and tuners** on the **Setup** tab; 4 to start) sets how many different Stations can play at once, across Plex and every other app.

- **One tuner per Station.** Everyone watching a Station shares one stream, however many watch.
- **When every tuner is busy,** someone tuning in to another Station sees an **All tuners in use** card for 30 seconds, listing what's on. Up to 3 people at a time see it.
- **Plex** is told there are more tuners than there are, so it lets tapiOca show that card. If you raise the number and Plex still won't play more, restart Plex.
- **Flipping back is instant.** A Station keeps running for 20 seconds after its last viewer leaves, and gives up its tuner if someone else needs it.

Test this server makes a few seconds of video at each size (up to a minute, at low priority) and says roughly how many Stations your server can play at once. A button such as **Use 4 tuners for 720p** fills in its suggestion, which leaves 30% headroom; then choose **Save**. 4K and HDR programs take more work, so treat it as a guide. The page warns you if you set more.

Schedules

Programs play back to back at their real lengths, never stretched to fill a time slot.

- **Episode order:** one episode of each show in turn, each show in order, like classic reruns. Shows go alphabetically, and a movie counts as a one-episode show. Then the run starts again.
- **Shuffle:** every program once a pass, each pass shuffled again, so a show airs at different times each day. No show plays more than twice in a row, unless one show makes up almost all of the Station: then it sometimes plays three or more times in a row, and the card says so. Add more shows to avoid that.

The schedule is worked out the same way every time, from when you saved the Station, so the guide stays right and a restart never changes what's on.

Updates to Stations

New episodes join Stations by themselves, and removed ones leave. Every hour, tapiOca prepares an update from what changed, and the card shows **Update ready**. An update never cuts a program short. It starts at a program break:

- **With Plex,** after Plex next downloads the guide (tapiOca asks it to), usually within the hour.
- **Without Plex,** once the guide other apps downloaded runs out, up to 2 days. If no app downloads it, at the next program break.

In **Shuffle**, new programs are spread through the pass, so 23 new episodes of one show don't become a marathon.

Updates that need your review. On a Station with 10 or more programs, an update that would remove more than half isn't applied by itself: it's more likely a drive offline than something you meant. The same goes when a Media folder suddenly reports nothing, or Plex loses most of the intro and credits markers. The card says "N programs gone — needs your review". Choose **Update now** if it's right.

Shows Plex adds again under a new ID are found again by title, and keep their places. If it can't tell which show is meant, the Station keeps what it had, and the **Logs** tab names the show to choose again. Specials (season 0) are left out of Stations.

Skipping intros and credits

With **Intros & credits** set to **Skip them**, each program airs without its opening titles and end credits, from the same markers as Plex's own **Skip Intro** and **Skip Credits**.

- **Cold opens still play,** and a scene after mid-program credits still plays.
- **No markers, no skipping.** Programs without markers play in full, including those from Media folders tapiOca reads itself.
- **Exact guide times,** without the skipped parts.
- **Only sensible markers:** an intro must start in the first half and be under 5 minutes; credits must start in the second half; at least a minute must be left.
- **Clean cuts.** tapiOca moves each skip to a change of shot or a moment of black near the marker, so a few seconds of titles may show (never more than 6 at each end). A skip only ever gets shorter.

Plex must find the markers first: under **Settings → Library**, set **Generate intro video markers** and **Generate credits video markers** to run as a scheduled task and when media is added (a Plex Pass feature).

Sound and picture shape

Even sound. TV episodes play at the same loudness (−24 LUFS, the US broadcast standard). Movies keep their original sound.

4:3 shows. Black bars keeps the original shape. **Stretch** widens the picture to fill the screen. **Zoom** fills it and trims the top and bottom. Widescreen programs are never changed. With **Stretch** or **Zoom**, black bars built into the picture are removed first.

The guide covers the next 2 days, in Plex, in tapiOca's apps, and under **Guide** on each card.

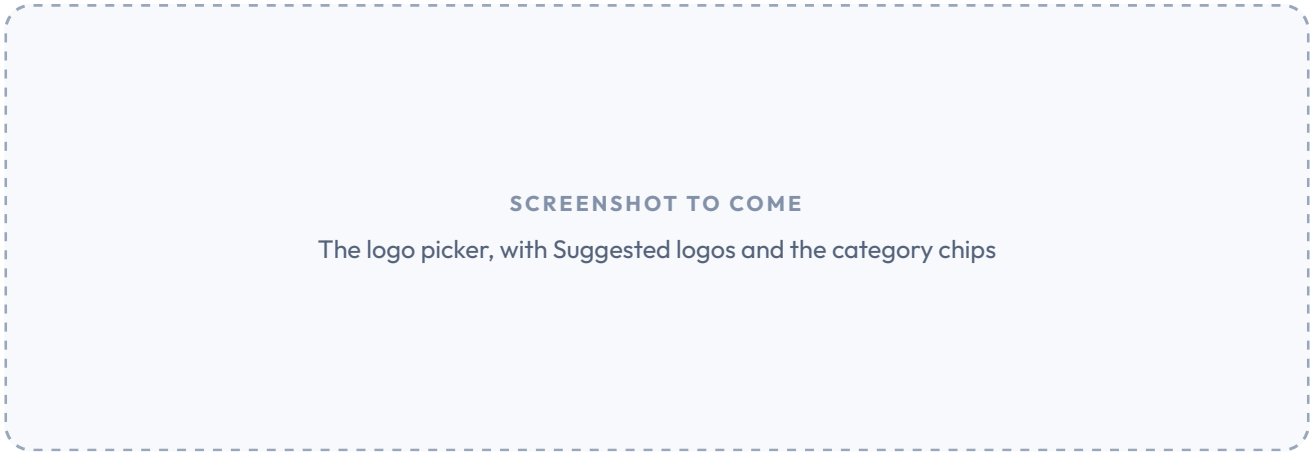
Station features

Each feature is set in a Station's editor, in the group named here. The Settings reference lists every choice and default.

Logos

tapiOca comes with 984 original logos, drawn for it. None copies a real network's or show's name, symbol or lettering. They're grouped as made-up networks and stations, styles (Classic TV, TV shows, Cartoons & anime, Teens, Kids & family), genres (three designs for each of Plex's 37), themes (decades, holidays, seasons, times of day, moods, on-air signs), and letters and numbers in a modern and a retro style.

- **Choose…** opens the logo picker, with search and categories. It opens on **Suggested** logos when the Station's name or filter suggests a genre. Faded logos are already used.
- **Surprise me** picks one at random. **Station number** shows the number instead.
- **Upload your own…** takes a PNG, JPEG, WebP, GIF or BMP image up to 10 MB, made into a 512×512 PNG (square images look best). Your logos are listed under **Your logos**, up to 500. An Admin can delete an unused one with its **×**.
- **Use the logo from Plex** takes the title artwork Plex has for the first show. A new Station with one show or movie is named after it, and gets its Plex logo by itself.



In the corner (On screen)

Shows the Station's **Logo**, its **Name** or a **Clock** in a corner during programs, never during commercials, trailers or cards. Set its **Size**, **Transparency**, **Corner** (or click a corner of the preview), the **Clock**'s format, the logo's **Look** (**In color** or **All white**), and **When** (**Always**, or **At the start**: the first 30 seconds of each program, and of whatever's on when someone tunes in).

Burn-in protection. So a still picture can't mark an OLED or plasma TV, what's in the corner steps every 4 minutes to the next of 9 places, all within 6 pixels. Every Station does it.

Up Next Banner (On screen)

Three minutes before each show or movie ends, a banner shows the Station's logo, **Up next** and the next program, for **3 sec**, **5 sec** or **10 sec** (or **Off**), at a **Size** you choose. **Preview** plays it. It takes the corner logo's place while it's up, and skips programs under 3 minutes.

Station ID card (Between programs)

After each program and its commercials, a card shows the Station's logo, name and number, and what's up next, for **3 sec**, **5 sec** or **10 sec**, with one of 15 original jingles (**Jingle On** or **Off**). **Preview** shows it. The card's time is part of each program's time in the guide.

Commercials and trailers (Between programs)

Commercials & trailers plays **None**, **1**, **2** or **3** clips after each program: commercials after episodes, trailers after movies. They're part of each program's time in the guide, each plays about as often as the others, and all play at an even loudness.

1. Make a folder named `commercials` at the top of your TV shows' Media folder, and `trailers` at the top of your movies' (Linux names are case-sensitive). Subfolders are fine.
2. Put clips from 3 seconds to 10 minutes long in them.
3. Put a `.plexignore` file containing `*` in each, so Plex doesn't add the clips:

```
echo '*' > /mnt/tank/media/tv/commercials/.plexignore echo '*' > /mnt/tank/media/movies/trailers/.plexignore
```

- **By decade.** Clips in a folder named for a decade (`commercials/1960s` or `commercials/60s`) play only after programs from that decade.
- **Under Viewing Levels.** On a Station someone with limits can see, a clip plays only if its folder is rated (see Viewing Levels and limits).
- **Finding them.** tapiOca looks at startup and every hour, or now with **Check again** in the editor. A clip that fails is left out until it changes, and listed under **Commercials and trailers that didn't play** on the **Broken files** tab.

Intro Bumper (When someone tunes in)

When a Station's stream starts, an Intro Bumper can play before the program:

- **Built-in card:** the Station's logo arriving, "You're tuning in to", its name, a one-line **Description** (up to 140 characters) and what's on, for 3 to 15 seconds, with the sound of an old TV dial (**Sound On** or **Off**). **Preview** shows it.

- **Your video: Upload a video...** up to 30 seconds long and 500 MB (MP4, MOV, MKV, WebM and similar). tapiOca makes a copy at the right size and loudness. Videos are kept in `data/bumpers` (not in backups), up to 50, and only Admins can **Delete** one.
- **Off.**

Someone joining while others already watch joins in progress, without a bumper.

Viewers join (When someone tunes in)

- **In progress:** like real TV. Tune in at 10:20 to a show that began at 10:00, and you join 20 minutes in. The show's sound fades in under the bumper.
- **From the beginning:** the program on now starts from its beginning, and the Station runs that far behind the guide while anyone is watching. A program that began more than 2 hours ago is joined in progress.

Subtitles (On screen)

Draws subtitles into the picture: **Off**, **Forced only** (lines meant to be read, such as foreign dialogue) or **Always**. They're in the `AUDIO_LANGUAGE` language (English unless set), from the file or a file beside it (`Movie (1999).en.srt`, `Movie (1999).en.forced.srt`). Viewers can't turn them off. A program's first airing may play without them while they're prepared.

Specials

Under **Specials**, three kinds of event take over a Station now and then. Each starts at the program break nearest its time, and the Station then picks up where it left off. A block wins over a Feature Presentation, and both over a marathon. A special more than 15 minutes late is skipped, and the **Logs** tab says so.

- **Marathons:** three episodes of one show in a row, at **Random times** (1 to 4 a week) or **Set times** (days and a time), playing **Next in order** or from a **Random starting point**. The Station needs at least 5 shows with 3 or more episodes each.
- **Feature Presentation:** a movie night on days and at a time you choose, after a 10-second card (**Preview the card**). **Movies from** is the Station's own movies, a movie Media folder or a movie collection, so a TV Station can have movie nights too. Every movie plays once before any repeats.
- **Blocks:** up to 4, such as Saturday Morning Cartoons. **Add a block**, then set its **Block name**, **Block days**, **Start time** and **End time** (30 minutes to 12 hours, past midnight is fine) and **Choose shows...**. A block ends at the program break nearest its end time, and picks up where it left off next time.

Saving is refused, with the reason, if specials clash or have nothing to play. The card shows when each is next.

Keeping Stations on the air

A bad file should never take a Station down. tapiOca retries, fills the time with a stand-in, checks files before they air, and lists what's wrong for you.

Safeguards

When	What tapiOca does
A program won't open, or stops partway	Tries again from where it stopped, twice.
It still won't play	Lists it on the Broken files list, and a stand-in joins at the same point and ends when it would have, so the guide stays right. A stand-in is another episode of the same show, then a program of the same kind, then anything long enough.
Opening a file is slow (a sleeping drive)	Keeps the stream alive with a few seconds of black.
A file stalls (no video for 8 seconds)	Resumes or replaces it. Stalls are blamed on storage: a file is listed only if it stalls in two separate viewings.
The corner logo, banner or subtitles fail	Plays the program again with nothing drawn over it before blaming the file.
Plex or the media share can't be reached	Skips those programs for now, without listing them.
Nothing on the Station can fill the time	Shows a "We'll be right back" card, then continues.
Nothing could play for two programs in a row	Shows "This station is off the air. Please report it.", keeps trying, and comes back by itself.
The Station's stream fails	Restarts it at once, so viewers see at most a brief pause. After 3 failures in 5 minutes, the off-air card shows for 2 minutes.

The Broken files tab

Files that can't play properly go on one list for Stations and Media alike, stored in `data/broken-files.json`. Each entry shows what's wrong, when it was found, how many times it failed, the file's path, the Stations that have it, and whether it's in Media.

- **On Stations,** every entry is skipped until it's cleared, and a stand-in plays instead.

- **In Media,** a broken file doesn't play (another version does, if there is one). A damaged one still plays.
- **Versions** of a program are separate files, each with its own entry.

Label	Meaning
Missing	Gone from disk or from Plex, while a Station or Media still has it.
Broken	Won't play.
Damaged	Plays, but the picture or sound breaks up.
Unsupported	Plays, but can't be shown correctly on Stations (Dolby Vision profile 5).

The tab has three parts. **Needs you** holds people's reports, and files tapiOca found that wait for your choice; the tab's count is these. With files of both kinds, choose what it shows: **All**, whose count is the tab's, with the reports; or **Broken or damaged** or **Missing**, which list files only. **Being replaced** is what Sonarr or Radarr is fetching now. **Found by tapiOca** is the rest: files you're handling, files nothing plays now, and Unsupported files. At the top, **Check the list again** goes through the list now, and **Download list** saves it as a file.

SCREENSHOT TO COME

The Broken files tab, with the file checks panel, Needs you and an entry's buttons

Clearing an entry yourself:

- **Retry** means "the file is fine. Put this program back on the air." The checks then leave that file alone. (An Unsupported file then plays with the wrong colors.)
- **Removed a show on purpose?** Take it off the Stations its entry lists, and its **Missing** entry clears within half an hour.
- **Deleting an entry** from `broken-files.json` also works (tapiOca notices within 15 seconds), but doesn't tell the checks to leave the file alone.

Entries that clear themselves: when Plex has the program again under a new ID, or a missing file is no longer on any Station or in Media (checked every half hour); when a new file for the program passes its quick check; and, for files a quick check took off the air, when they pass again overnight. Deep-scan and playback problems stay until the file changes.

File checks

tapiOca checks files at the lowest priority, to find problems before anyone tunes in, and steps aside while anyone is watching. Each file is checked once for Stations and Media together. In order:

1. **Programs airing soon** (in the next 3 hours) that are due a check.
2. **A file someone just had trouble with:** tapiOca decodes the minute around where it happened, then quick-checks the file. Only what tapiOca finds lists it, so a network hiccup changes nothing.
3. **New arrivals,** quick-checked within minutes: the file must open, and its picture and sound must decode at five points.
4. **Weekly,** every program on a Station is quick-checked again.
5. **The overnight deep scan** decodes files in full, Stations' programs first, then Media's (Resume rows and next episodes first). It stops the moment anyone starts watching. Each file is scanned once, and again only if it changes.

Deep scan every night from, on the **Broken files** tab, sets its start and end times (1:00 AM to 6:00 AM to start), then **Save**. Untick it to turn the deep scan off. **Check files** on a Station card quick-checks all its programs now. The panel at the top of the tab shows how far the checks have gotten.

Result	What it means
Broken	Won't open, has no picture, stops decoding, ends early (more than 10 seconds or 2% short), or can't be read on two different nights.
Damaged	The picture breaks up, the sound drops out (20 ms or more lost within a second), or the file skips; no sound, or nothing, for 30 seconds or more; no sound track or silence throughout (for programs from 1930 on); an all-black picture; sound that stops before the picture; or a file that runs well past its stated length.
Fine	Warnings no one would see or hear, anything in the first 2 seconds or last 5, pauses under 30 seconds, a still picture over sound, a black opening, or a file slightly short.

Each entry says what was found and where, such as "the picture breaks up around 12:31 and 48:02". Times are approximate.

Replacing files with Sonarr and Radarr

tapiOca doesn't need them. But if you use **Sonarr** (TV) or **Radarr** (movies), version 3 or later, tapiOca can ask them to replace files on the list that a Station or Media plays.

1. On the **Broken files** tab, open **Replacing files with Sonarr and Radarr**.
2. Tick **Use Sonarr for episodes** (or **Use Radarr for movies**).
3. Enter its address, such as `http://192.168.1.10:8989`, and its **API key** (in the app under **Settings → General**).

4. Choose **Test**, then **Save**. (An address on your home network asks first.)
5. Choose **Files to replace** (**Broken or damaged**, **Missing** or **Both**) and **When to replace** (**Only when I ask** or **Automatically**).

With **Only when I ask**, choose **Replace with Sonarr** (or **Radarr**) on an entry. With **Automatically**, every entry is replaced, unless you choose **I'll handle it** on it. API keys are stored in tapiOca's database and backups, and never shown again.

What happens. For a broken or damaged file, the app first **blocklists** the release it came from, then deletes the file and searches for another. This happens only if the app's file is exactly the one tapiOca found, and the app has a record of fetching it. For a missing file, the app rescans and searches. Each gets up to 3 searches, 8 hours apart. tapiOca checks each new file, and puts the program back on the air once it passes. It only asks about what the app is monitoring: unmonitor something there to leave it alone.

The same problem twice may be part of the program itself, such as an old film's transitions. tapiOca stops and marks the entry **Needs your review**. Watch it there: choose **Retry** if it's fine, or **Try another file**.

Each entry shows **Replacing**, **Downloading**, **Downloaded**, **Gave up** (3 searches found nothing; **Try again** starts over), **Can't replace**, **Needs your review** or **You're handling it**. The **Logs** tab records every step. Unsupported files aren't replaced this way: you're the best judge of the right version.

People's reports

People can report a problem from tapiOca's apps, from a title's page, the player's menu, or a Station's player. They pick it from a list: **Picture** (No picture; The picture breaks up or freezes; Poor picture quality), **Sound** (No sound; The sound cuts out; The sound is out of sync; Wrong language), **Subtitles** (Subtitles are missing or wrong), **The program** (It won't play; It stops before the end; Wrong episode or movie) and **Details** (Wrong title, details or artwork).

Reports go under **Needs you**, with who sent it, from what device, where in the program, and how it was playing.

- **What tapiOca can check** (no picture, breaking up, no sound, cutting out, stopping early, not playing): it checks the file at once. If it finds the problem, the file is listed. If not, the report says so, for you to **Dismiss**.
- **What only a person can judge** (sync, language, wrong episode, picture quality, subtitles): it waits for you, with what tapiOca can tell about the file. Choose **Replace with Sonarr** (or **Radarr**), **Find a better copy** (an upgrade, keeping the file meanwhile) or **Dismiss**.
- **Wrong title, details or artwork:** fix it in Plex (**Fix Match** or **Edit**), then **Dismiss**.

A report never takes anything off the air by itself. Each person can send one report a day about each program, and 10 a day in all. Turn reporting off for someone with **Can report problems**, on the **Access** tab or under **Reports from tapiOca's apps** here. Reports are kept 90 days after they're handled.

People, sign-in and Viewing Levels

Everyone who uses tapiOca's page or apps signs in as themselves: someone an Admin added on the **Access** tab. Signing in doesn't change who can watch in Plex, Jellyfin, Emby or Kodi: each app decides that itself.

Roles

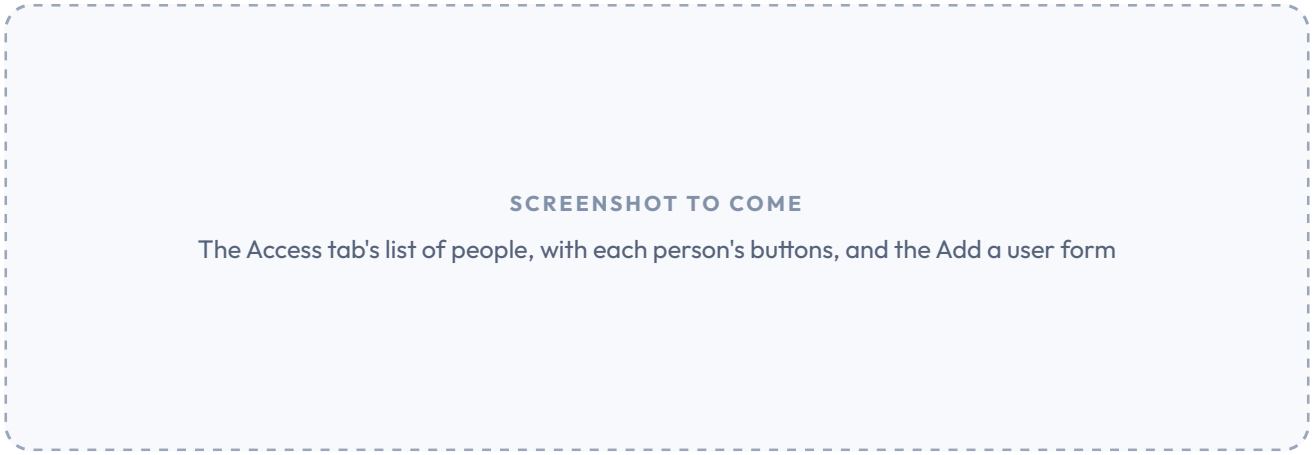
Role	Can do
Admin	Everything: every Station, and the **Setup**, **Access**, **Logs** and **Broken files** tabs.
User	Watch what their Viewing Level allows, see **Stats**, make a few Stations (3 to start) and change only their own, and add logos and Intro Bumper videos. On a Viewing Level with limits, they only watch.

The first person added is always an Admin, and there's always at least one. Until there's an Admin, the page and apps are open to anyone at home, and closed to the internet.

> **Important:** Skipping the first-time setup on a new server leaves no Admin, so anyone at home can change anything. Add one right away.

Adding a person

1. On the **Access** tab, under **Add a user**, enter a **Name** (up to 40 characters, unlike anyone else's).
2. Enter a **Password**, optional for a User.
3. Optionally, a 4-digit **PIN (optional)**.
4. Choose a **Role**, and for a User, a **Viewing Level** and **Stations they can make**.
5. Choose **Add**.



Each row shows the person's **Name**, **Role**, **Viewing Level**, **Stations they make** and **Last signed in** (when, and from what address). Beside each:

- **What they see:** Stations and titles for this person (see Viewing Levels and limits).
- **Rename:** they keep everything, and sign in with the new name.
- **Devices:** the devices they're on, their PIN, and **Make an invite code**.
- **New password** (or **Add a password**): signs them out everywhere else.
- **Set PIN** (or **Change PIN**), and **Remove**.

Three permissions under each person start on: **Can change their own password**, **Can change their own PIN** and **Can report problems**. Admins always can. A new role ends all of that person's sign-ins, but the one making the change.

Passwords and limits

- **Passwords** need at least 8 characters, with no rules about symbols or capitals. tapiOca's developer chose this, and it's less than NIST's guidelines (SP 800-63B) ask for, 15 characters. A longer one is safer: a few words together work well. tapiOca turns away common passwords (checked on the server against a list it keeps), repeats and runs, and ones built from the person's name or tapiOca's. Every password box has **Show**.
- **Passwords and PINs are stored only as salted hashes.**
- **A sign-in lasts 30 days** after it was last used. The page signs you out after an hour without activity.
- **Wrong passwords:** after 5 from one address in 15 minutes, that address waits.
- **Wrong PINs:** after 5, that person waits 15 minutes, on every device. After 10 on one browser, it needs a password.
- **No password.** A User may have none, such as a "Kids" user who only picks themselves on the TV. They get on a device with an invite code.

Full or Light sign-in

Each Admin chooses, under **How you sign in** in **Your account**:

- **Full** (the default): your password every time you sign in for Admin tasks.
- **Light:** your password once on each browser or device, then your PIN there, at home. It needs a PIN.

Through tapiOca's internet address, the page never takes a PIN. A PIN's sign-in can't change passwords or PINs, add people, make API tokens or invite codes, or link an app. Choosing **Full**, or a new PIN, ends the sign-ins the old PIN made. A User's PIN never signs in on the page.

Linked devices and Who's tuning in?

The first time someone signs in on a device, such as the living room TV, it's **linked**: from then on it opens on **Who's tuning in?**, and whoever is watching picks themselves.

- **Who's on the list:** only people who signed in on that device themselves, with their password, an invite code, or **Link an app**. They're on it at home and away. Someone with no password and no PIN is on it only at home.
- **PINs.** A person with a PIN gives it when they pick themselves. The app asks for a PIN, or none, the first time they sign in. An Admin can give their password instead, and gets Admin tasks from a PIN only in Light sign-in, at home.
- **Taking yourself off. Sign out** in an app takes you off that device's list, and ends your sign-ins there, even if your sign-in there had already ended. One exception: if the app went back to **Who's tuning in?** by itself before you pressed **Sign out**, you're still listed. Pick yourself, then **Sign out**, or have an Admin **Unlink** the device.
- **A sign-in from Who's tuning in?** lasts a day from when it was last used, so a shared device returns to the list.

> **Important:** On a shared device, anyone can pick anyone who has no PIN. Give everyone without limits a PIN on devices that people with limits use.

Invite codes. Choose **Devices** beside a person, then **Make an invite code**. On the device, choose **Sign in** on **Who's tuning in?**, and enter their name and the code. A code works once, for 7 days.

Signed-in apps lists each app, as whom, and when it was last used: **Sign out** ends one at once, and one unused for 30 days ends by itself. **Linked devices** lists each shared device and who's on it: **Unlink** signs it out.

The access log and lockouts

On the **Logs** tab, tick **Access** to see sign-ins, failed sign-ins, sign-outs and changes to who can sign in. The newest 2,000 entries are kept across restarts.

Locked out? Create an empty file named `reset-access` in tapiOca's data folder (`sudo touch data/reset-access`), and restart tapiOca. Every person, every trusted network that signs in a person, and every trusted proxy is removed, and sign-in turns off. The file is deleted. Add an Admin again right away.

Viewing Levels and limits

Each person has a **Viewing Level** that sets what they can see in tapiOca's apps and on its page. Admins always see everything. The server decides every limit, and the apps show only what it tells them.

Viewing Levels

On the **Access** tab, open **Viewing Levels**. Choose each person's level beside their name, or as you add them.

Level	Movies up to	TV up to	Unrated titles
Unrestricted (can't be changed)	No limit	No limit	Shown
Teen	PG-13	TV-14	Hidden
Kid	PG	TV-PG	Hidden
Young Child	G	TV-G	Hidden

New people start on is the level new Users get: **Unrestricted** until you choose another.

To change a level, choose **Edit** beside it (or **Add a level** for your own, such as "Grandparents"), set these, then **Save**:

- **Name**, **Movies up to**, **TV up to**, and **Unrated titles** (**Shown** or **Hidden**).
- **Media folders:** all, or only some. A show in two Media folders is then seen only through the one the level includes.
- **Genres**, **Tags**, **Collections** and **Studios:** type a name and choose **Add** to hide anything with it, whatever its rating. Plex's names are offered as you type. Tags are what Plex calls labels.

Ratings are read as ages, for the US and other countries alike. An episode counts as its show, or its own rating if that's stricter. A title with no rating, or one tapiOca doesn't know, is unrated. To give one person ratings of their own, give them a level of their own.

SCREENSHOT TO COME

The Viewing Levels panel, with the four built-in levels and Also hides

What they see

Choose **What they see** next to someone's name. For Kit, **What Kit sees** has three parts:

- **Stations:** for each, **As Kid says** (their level), **Allowed** or **Blocked**, with why each is shown or hidden.

- **Also hidden for Kit:** genres, tags, collections and studios, and titles you find by typing, hidden whatever their level shows.
- **Allowed for Kit:** titles their level would hide, shown anyway, if they're in a Media folder the level includes.

How limits work on Stations

Everyone watching a Station sees the same stream, so:

- **A Station is shown** to someone only if everything on it is within their level. Allow or block it under **What they see** to change that.
- **"Restricted program."** On any Station, a program someone may not watch is blocked for them while it airs, and their guide shows "Restricted program". Their Station comes back by itself afterward, and everyone else watches on.
- **Limits are checked while a title plays,** so a change stops it within seconds.
- **Nothing hidden shows** anywhere for them: not in Media, search, **More like this**, cast pages, the Resume row or the guide.
- **People with limits only watch:** they can't make Stations.

Until tapiOca knows a Station's ratings, or a title's genres and tags, it's hidden from anyone it might not suit.

Commercials and trailers

On a Station someone with limits can see, a clip plays only if its folder is rated at or below the Station's highest-rated program. Under **Commercials and trailers**, below the levels, rate **Commercials** (TV ratings), **Trailers** (movie ratings) and each decade's folder, then choose **Save ratings**. Until then, those Stations skip unrated clips, and an alert says how many.

What tapiOca enforces, and what it can't

Viewing Levels work in tapiOca's apps and on its page, for someone signed in as themselves. They don't cover:

- **Plex, Jellyfin and anything else on the tuner,** or a Station's stream address played at home (such as in VLC). They don't say who's watching, so they show everything. See Tuner access, and Blocking Stations for Plex users.
- **Devices without PINs,** where anyone can pick anyone.
- **Someone who knows another's password.**
- **A rating or genre Plex has wrong,** or what's inside a program beyond its rating, genres and tags.

> **Tip:** Use your devices' own controls too, such as Apple's Screen Time or Google's Family Link.

The access log records each change to a level, a person's choices and the clip ratings.

Blocking Stations for Plex users

Plex shows every Station to everyone with Live TV on your server. With Plex Pass on the server owner's account, tapiOca can stop a Plex user's playback instead.

1. On the **Access** tab, open **Block stations for some Plex users**.
2. Open each Plex user and tick the Stations they can't watch.
3. Tick **Stop Plex users from watching stations blocked for them**, and choose **Save**.

When a blocked user tunes in, Plex stops it within seconds and shows "This station isn't available on your Plex account." The panel lists **Recently stopped**. This covers Plex apps only (you, Plex Home users and friends you've shared Live TV with), needs a Plex user for each person, and needs sign-in on in tapiOca. If two Stations air the same program at once and Plex gives no Station number, tapiOca can't tell them apart, and stops neither.

Home, away and your network

tapiOca treats every connection as either at home or from the internet. Several panels on the **Access** tab decide what each may do.

At home or from the internet

Anything reaching the public port (3311) is from the internet, even from inside your home. A connection to the home port (3310) is at home only when it comes from a home or VPN address (such as 192.168.x.x, 10.x.x.x, 172.16.x.x to 172.31.x.x, or Tailscale's 100.64.x.x), doesn't ask for tapiOca by an internet address, and no proxy says it's from the internet. Anything else is treated as from the internet, and gets only what the public port gives.

Home names. Trusted networks also need tapiOca asked for by a home name: an address, a single word (such as `nas`), a name ending in `.local`, `.lan`, `.home`, `.home.arpa`, `.internal`, `.localdomain` or `.ts.net`, `BASE_URL`'s name, or one listed under **Names for tapiOca at home**. This also stops another web page from pointing a name of its own at tapiOca and using your browser to change it (DNS rebinding).

Trusted networks

Devices on a trusted network skip signing in. Nothing is trusted until an Admin adds it.

1. On the **Access** tab, open **Trusted networks**. It says where you are, such as "You're at 192.168.1.44, on Home Wi-Fi."
2. Enter a **Network, or one device's IP address** (such as `192.168.1.0/24`, `192.168.1.44` or your VPN's `10.8.0.0/24`), or choose **Add the network you're on**.
3. Give it a **Name (optional)**, such as "Home Wi-Fi", and choose what **Its devices** get: - **Who's tuning in?** (the default): in tapiOca's apps, the device is linked without signing in, and opens on **Who's tuning in?** Its list starts empty: each person chooses **Sign in** there once. - **Signed in as** a person (such as **Signed in as Ada**): the device is signed in as that person, for watching, in the apps and on the page. For an Admin, only one device's address: reserve it in your router.
4. Choose **Add**.

> **Important:** Never add a guest network. Anyone on a trusted network can use tapiOca's apps without a password.

- **Home addresses only,** up to 50. A network wider than 256 addresses asks first. The narrower of two overlapping entries wins.
- **Only there.** What a trusted network signs in works only on that network, never away from home. Changing or removing the network ends it.

- **For watching.** A trusted network's sign-in can't change passwords or PINs, add people, make tokens or invite codes, or link an app. An Admin gives their password for Admin tasks.
- **Only when tapiOca can be sure:** signing in on, a home name, and tapiOca able to see where connections come from. Nothing is ever trusted through Docker's own address.

Trusted proxies

List your reverse proxy or tunnel under **Trusted proxies**, by its **Proxy's IP address** on your home network, then **Add**. List up to 10 (or `127.0.0.1` for a tunnel in tapiOca's own container).

- **Only a listed proxy is believed** about its visitors, in `X-Real-IP`, `X-Forwarded-For` and `Forwarded`, which must all name the same visitor. Headers from anything else are ignored, and Admins get an alert.
- **HTTPS.** tapiOca believes a request came over HTTPS only from a listed proxy, so the apps from the internet need their proxy listed.
- **Trust this proxy.** When tapiOca's away-from-home check comes through a proxy that isn't listed, it offers **Trust this proxy**, which checks again and lists just that address.
- **A proxy in another container on the same server** reaches tapiOca from Docker's own address (such as `172.17.0.1`). List it so wrong passwords count by each visitor, but nothing is trusted through it. For trusted networks through it, give the proxy its own address on a Docker network shared with tapiOca.

Names for tapiOca at home

If you open tapiOca at home by a name such as `tapioca.example.com` (split DNS), list it under **Names for tapiOca at home** (up to 10), so trusted networks work by it. Your away-from-home address's name can't be one.

Doors left open

When tapiOca can't be sure a connection is from home, it treats it as from the internet, and alerts Admins (never with an address or a name):

Left open	What tapiOca does
The home port forwarded on the router	Treats it as from the internet. Close the forward.
A reverse proxy or tunnel pointed at 3310	Treats its visitors as from the internet. Point it at 3311.
An internet name on 3310	Never trusts it. If it's a home name, list it under **Names for tapiOca at home**.
Forwarding headers from a device that isn't listed	Ignores them. If it's your proxy, list it.
Docker hiding every device's address	Turns trusted networks off.

Left open	What tapiOca does
Signing in left off	Open at home only. Nothing from the internet gets in.
An internet address asking for the tuner	Never gives it the tuner.

When Docker hides addresses. Some Docker setups pass every connection through Docker's own address, as Docker Desktop always does. When that happens, trusted networks turn off, and the **Access** tab says why. The fix: host networking (`network_mode: host`, in place of the `ports:` lines), or `"userland-proxy": false` in Docker's `daemon.json`, then restart Docker and tapiOca. On Docker Desktop there's no fix: every device signs in. When only some connections are hidden, open tapiOca by its IPv4 address.

Tuner access

The **tuner** is what Plex, Jellyfin and other apps use: the HDHomeRun tuner, the guide, the list of Stations and their streams. None of these apps can sign in. It's on the home port only. Under **Tuner access** on the **Access** tab, choose who can use it:

- **Any device on my home network** (the default).
- **Only these devices:** enter a **Device's IP address, or a network** and a **Name (optional)**, then **Add** (home addresses, up to 50). **Plex server only** lists just Plex's address, after showing you which it is.

Recent tuner connections shows each device that used the tuner in the last 7 days, what it asked for, and whether it was **Refused**. **Allow** lists one. Plex in Docker often comes from the Docker network's address, so use **Plex server only** or **Allow** rather than typing the server's address. Any device can call itself Plex: check the address first.

> **Note:** An IP address isn't a password. Tuner access keeps out devices that find the tuner by mistake, not someone at home who means to get in. tapiOca's own apps, signed in, never need listing. Turn sign-in on first, so no one else can change this.

Away from home

Often you don't need a way in. Plex users away from home watch through Plex, which relays the stream. Only tapiOca's page and apps need a way in from outside.

> **Important:** Never open port 3310 to the internet. Its tuner and streams never ask for a password. Every way in below keeps it closed.

Way in	Good for	Open to the internet	Cost
Tailscale (a VPN)	The page and apps on your own devices	Nothing	Free for personal use
WireGuard on your router (a VPN)	The same	One port, for the VPN	Free
A reverse proxy with HTTPS	The apps anywhere, for anyone you've added, with no VPN; also the page	Port 443, to the proxy	A domain name, about $10 to $15 a year
A Cloudflare Tunnel	The page only	Nothing	Free

A VPN is the most private, and tapiOca treats a device on it as at home. A reverse proxy suits people who won't install a VPN. A Roku can't run a VPN, so it needs the reverse proxy.

Turning on the apps away from home

1. At home, add the first Admin, with a long password, if you haven't.
2. On the **Access** tab, open **tapiOca's apps away from home**, and tick **Let tapiOca's apps watch away from home** (off to start).
3. Under **The address apps reach tapiOca at from outside**, enter your reverse proxy's `https://` address with no port (such as `https://tv.example.com`), or your VPN address with its port (such as `http://nas.your-tailnet.ts.net:3310`).
4. Choose **Media away from home: Original** (the default, as at home), or **Up to...** 1 to 200 Mbps, which converts larger files down to fit, at most 1080p.
5. Choose **Save**. The status below should show **Ready**.

Each Station watched away from home uses your upload: about 1.7 Mbps at 480p, 3.7 at 720p and 6.2 at 1080p. The panel shows your upload as the apps measured it from outside. Without `PUBLIC_PORT`, apps reach tapiOca from outside only through a VPN.

SCREENSHOT TO COME

The tapiOca's apps away from home panel, with an https address and the Ready status

The reach check. While it's on, tapiOca calls its own address as an app away would: at startup, when you save, every 5 minutes, and with **Check now**. **Ready** means it works. Anything else says the problem in one sentence, such as a proxy pointed at 3310, no HTTPS, a name that isn't found, a bad certificate, or a redirect. **Away from home** in the header shows **Up**, **Down**, **Checking** or **Can't check**. One failure never makes it **Down**.

When your router answers from home. Many routers don't let devices at home reach the home's own internet address (NAT loopback, or hairpinning). Apps away still work, but tapiOca's check from home can't, so it shows **Can't check**, or **Up** when an app came in from outside in the last 15 minutes. To be sure, open your address on a phone using mobile data. To fix it, add a DNS host override in your router (such as Unbound's, in OPNsense or pfSense) pointing your address's name at the proxy's home address.

Tailscale

1. Install Tailscale on the server (on TrueNAS, **Apps → Discover Apps → Tailscale**; on Unraid, its plugin; on Synology, **Package Center**), sign in, and turn on **MagicDNS**.
2. Install it on each device that should reach tapiOca, on the same account.
3. Use the server's Tailscale name with port 3310, such as `http://nas.your-tailnet.ts.net:3310`, for the page and as the apps' outside address.

WireGuard on your router

Turn on your router's built-in WireGuard server, and add each phone or tablet in the WireGuard app. While connected, use tapiOca's home address as if you were home.

A reverse proxy with HTTPS

The proxy (such as Caddy or Nginx Proxy Manager) takes HTTPS from the internet and passes it to tapiOca's **public port**, 3311, never 3310. You need a domain name pointing at your home (with dynamic DNS if your address changes), port 443 forwarded to the proxy, and a certificate.

Caddy (the whole `Caddyfile`):

```
tv.example.com {
    reverse_proxy 192.168.1.20:3311 {
        header_up X-Real-IP {remote_host}
    }
}
```

Nginx Proxy Manager (or NPMplus): a proxy host for `tv.example.com`, forwarding to `192.168.1.20`, port `3311`, with a Let's Encrypt certificate and **Force SSL** on its **SSL** tab.

Then, at home, list the proxy under **Trusted proxies**, and turn on the apps away from home with its `https://` address. The proxy must pass each visitor's address in `X-Real-IP` or `X-Forwarded-For`, replacing any a visitor sends.

On the public port, apps connect only over HTTPS; each app's stream addresses are its own and end with its sign-in; and API tokens are refused unless an Admin allows them.

A Cloudflare Tunnel (the page only)

A tunnel reaches the page without opening a port. It's not for the apps: Cloudflare's free plan isn't meant for video.

1. At home, add the first Admin, with a long password. Keep `PUBLIC_PORT` and its port line.
2. In Cloudflare **Zero Trust**, add a public hostname to your tunnel with the service `http://<your-server-ip>:3311`. Never 3310.
3. List the address `cloudflared` reaches tapiOca from under **Trusted proxies** (`127.0.0.1` if it runs in tapiOca's own container).
4. **Strongly recommended:** in Zero Trust **Access**, add the hostname as a self-hosted application allowing only your people's email addresses, so Cloudflare asks for an email code first.

Watching in other apps

Your Stations work in other apps on your home network: as a tuner and guide for Plex, and as a playlist and guide for Jellyfin, Emby, Kodi and others. Everyone watching a Station shares one stream.

For	Addresses
Plex (as an HDHomeRun tuner)	Tuner `http://<server-ip>:3310`, guide `http://<server-ip>:3310/xmltv.xml`
Jellyfin, Emby, Kodi and others	Playlist `http://<server-ip>:3310/stations.m3u`, guide `http://<server-ip>:3310/guide.xml`
Any HLS player	`http://<server-ip>:3310/hls/<station number>/index.m3u8`

Tuner integration, at the top of the **Setup** tab, shows these with **Copy** buttons, and three status lines: **Media files** (whether Plex's files are read from disk), **Video encoding** (GPU or processor, and why) and **Guide updates** (whether tapiOca can ask Plex to refresh its guide). Opened from the internet, it shows the address Plex uses for tapiOca's tuner, or `http://<your NAS>:3310` until Plex has one.

> **Important:** These addresses work at home only, and never ask for a password. Never forward port 3310. Plex users away from home still watch through Plex, which relays the stream.

SCREENSHOT TO COME

The Setup tab's Tuner integration panel, with its addresses and status lines

In Plex

1. In Plex, open **Settings → Live TV & DVR → Set Up Plex DVR**.
2. Choose **Don't see your HDHomeRun device? Enter its network address manually**, and enter the tuner address. Plex doesn't find tapiOca by itself.
3. When Plex asks for guide data, choose **Have an XMLTV guide on your server? Click here to use it**, and enter the guide address for Plex.

4. Continue. Plex matches your Stations (it calls them channels), and they appear under **Live TV**.

tapiOca then asks Plex to refresh its guide whenever a Station changes. If **Guide updates** says it can't, use **Settings → Live TV & DVR →** your tuner **→ Refresh Guide**. After you make a new Station, choose **Scan for channels** there. A Plex Home or managed user needs **Allow Live TV and DVR access** to see Stations. Plex won't add a tuner with an empty guide: the INFO Station fills it until you make one.

In Jellyfin, Emby and Kodi

- **Jellyfin:** in the dashboard, open **Live TV**. Choose **+** next to **Tuner Devices**, then **M3U Tuner**, with the playlist address. Then **+** next to **TV Guide Data Providers**, then **XMLTV**, with the guide address. Leave Jellyfin's limit on simultaneous streams at no limit: tapiOca enforces its own.
- **Emby** (needs Emby Premiere): in **Live TV**, add a TV source, **M3U Tuner**, with the playlist address. Then add a guide data provider, **XMLTV**, with the guide address.
- **Kodi:** install the PVR Simple Client add-on (**Add-ons → Install from repository → PVR clients**). In its settings, set the playlist location to remote and **M3U play list URL** to the playlist address (it tells Kodi where the guide is). Restart Kodi. Your Stations appear under **TV**.
- **Other apps,** such as Channels DVR and VLC, take the playlist address.

These apps download the guide on their own schedule, about once a day. After you change a Station, refresh the guide in the app (in Jellyfin, **Dashboard → Scheduled Tasks → Refresh Guide**).

HLS shares the Station's one stream, so it never takes a second tuner. It runs only while a player asks for it, and stops 30 seconds after the last one stops.

What other apps don't get

Other apps on the tuner get your Stations, and nothing else. Media on demand, Viewing Levels, **Who's tuning in?**, languages that follow each person, and problem reports work only between tapiOca's server and its apps. Other apps don't say who's watching, so they show every Station and program. See Tuner access and Blocking Stations for Plex users.

tapiOca's own apps

tapiOca's apps connect straight to your server, so people can watch your Stations and Media without Plex Pass or any other app. They're in beta, in private testing: tapiOca for Android (phones, foldables, tablets, Google TV and Android TV), for iPhone and iPad, and for Roku. This chapter covers the server's side. The manual for tapiOca's apps covers each app.

> **Note:** Media on demand, Viewing Levels, **Who's tuning in?**, languages that follow each person, and problem reports work only between tapiOca's server and its apps. That's on purpose: they're built together.

Which server the apps need

The apps need tapiOca 1.22.0 or newer, and browse fastest with 1.43.0 or newer. An app on an older server says so. Newer servers add: PINs (1.28.0); languages and Admin alerts (1.29.0); converted copies (1.29.1); reporting a problem (1.30.0); automatic subtitles (1.34.0); trusted networks (1.36.0); **Shuffle all**, **Shuffle unwatched** and jumping through a Media folder (1.37.0); invite codes and Admin passwords on **Who's tuning in?** (1.38.0); "Restricted program" (1.39.0); limits checked while a title plays (1.40.0); and faster Media folders (1.43.0). Use the newest server for every feature.

Connecting an app

Add the first Admin before you connect the apps. Until there's an Admin, the server lets anyone at home use them without signing in, and they all share one Resume row.

1. **Find the server.** At home, an app can look for tapiOca on your network, or you enter the address you use for tapiOca's page at home, such as `http://192.168.1.20:3310`.
2. **Sign in:** with a code on a TV (entered with **Link an app** on tapiOca's page), with a name and password, with an invite code, or with nothing at all on a trusted network.
3. **Choose a PIN,** or none, when the app asks.

Each signed-in app appears under **Signed-in apps** on the **Access** tab. Apps set up at home remember the away-from-home address, and switch to it when home doesn't answer.

Media in tapiOca's apps

On the **Access** tab, open **Media in tapiOca's apps** (also in the setup). Tick the Media folders the apps can show, choose **When playing can't keep up** for **At home:** and **Away from home:**, and set **Even sound for a show's episodes**. Then choose **Save**.

SCREENSHOT TO COME

The Access tab's Media in tapiOca's apps panel

- **The best picture and sound.** A file plays as it is whenever the device can play it, 4K, HDR, Dolby Vision and Dolby Atmos included.
- **Copies only when needed.** Otherwise tapiOca makes a copy as it plays. A **repackaged** copy keeps the picture and converts only the sound, which costs almost nothing. A **converted** copy makes the picture again, at up to 1080p in standard color: at most 3 at once, or 6 with a GPU. Copies also draw in subtitles a device can't show, and make smaller pictures for slow connections.
- **One listing per title,** whatever its versions. Each device plays the best one it can. A movie in several files plays as one.
- **When playing can't keep up,** after the buffer, a short test finds out why, and the app **Stop, and offer a smaller version** (at home, to start) or does **Switch to a smaller version on its own** (away, to start).
- **Each person's own place:** their Resume row, resume points and watched list.
- **Even sound for a show's episodes** (on to start) plays episodes at −24 LUFS, as on Stations. Movies aren't changed. The cost: Dolby Atmos or DTS sent on to a receiver comes as ordinary 5.1 or stereo. Turn it off to keep each episode's sound as it is.
- **Languages.** Each person chooses sound and subtitles in the app's Options, and the choice follows them to every device. Subtitles start **Automatic**: shown when the sound isn't in their language. They can also choose for one title, or a whole show with **Sticky Audio & Subtitles**. Stations use `AUDIO_LANGUAGE` instead.
- **Night mode** on devices that can't change the sound themselves (Apple's player, a Roku) is made on the server, only for those watching that way.

How many can watch at once

Each device watching in the apps gets its own copy of the stream, over your network at home and over your upload away. So these limits are about your network, not your processor. On the **Access** tab, open **How many can watch at once in tapiOca's apps**:

- **Devices watching at once:** every device, at home and away, Media included.
- **Away from home, at most:** how many of those can be away, so your upload doesn't run out.

Leave a box empty for no limit (the default), then **Save**. A device already watching can always change Stations. One beyond a limit is told an Admin has limited tapiOca, and to try again later. Plex and other apps aren't counted here.

What tapiOca recommends comes from connection tests that time tapiOca's own data (never an outside service), using the fastest recent test with 30% headroom. **Use the recommended numbers** fills them in. **Test the connection** runs a test from your browser, counted as at home or away by where the page is opened. To test away from home, run it from a phone on mobile data, in an app's Options. **Recent connection tests** lists the latest.

The apps' problems show on the **Logs** tab, and people's reports on the **Broken files** tab. Admins see tapiOca's alerts in the apps too.

Stats, logs and alerts

The Stats tab

Everyone can open **Stats**. Choose **Today** (since midnight), **7 days** (the default), **30 days** or **All**. It shows totals; for each Station its **Viewings**, hours **Watched**, **Average**, **Last watched** and **Most watched** title; the most watched shows and movies; and **When people watch**, by time of day.

A viewing counts once it lasts a minute. Viewings are counted per stream, so several people watching one Station together in Plex can count as one. Stats are kept for 400 days, and a Station's go with it.

Admins also see:

- **Server** (marked **Live**, every 5 seconds, with 10-minute charts): **Processor** and **Memory** (tapiOca's and the machine's, or the **Container's limit**), **Sending** and **Receiving**, the **GPU** in use, and the data folder's **Storage**.
- **Watching now: Who**, what, the **App and device**, at home or away, **Since** when, and **How** it's sent (**As it is**, **Repackaged** or **Converted**, its size and bitrate).
- **Top people:** who watched most, and what. In the apps, viewing counts for whoever is signed in. For Plex, tapiOca asks Plex every 30 seconds who's watching each Station, and marks Plex users **Plex**. Other apps aren't counted per person.
- **Top Media:** the titles played most in Media.

SCREENSHOT TO COME

The Stats tab, with the Server tiles, Watching now and the Station table

The Logs tab

The **Logs** tab shows the latest 200 entries, newest at the bottom. **Search the log** finds words through everything it keeps (its newest 500 lines, and with **Access** ticked, the newest 2,000 sign-ins and changes). **Warnings** and **Errors** filter it. **Access** shows the access log. **Copy** copies what's shown. The activity log starts again when tapiOca restarts. Its lines also go to the container's output (`docker logs stationplay`), which Docker keeps until it rotates it: the install files keep 3 files of 10 MB each.

The log names each Station ("station 2, Cartoon Classics") and says what plays and how: a program's file, picture and sound, and how the Station makes it ("made 720p at 3.5 Mbps on the Intel/AMD GPU"); who started what in Media, and whether it plays as it is, repackaged or converted, and why; steps down in quality; and who stopped what, and when.

Problems in the apps, at the top, lists what the apps report as they play: a Station that doesn't start or stops, Media that doesn't play, playing that can't keep up, the app closing, or the app not reaching tapiOca (told once it's back). Choose **Today**, **7 days** or **30 days**. Each problem shows how often, on how many devices, for whom, and on what kinds of device. **One kind of device** points to that device or its app. **What it means**, **What the app said** and **What led up to it** explain each one. Problems are kept for 30 days. **Clear** removes them.

SCREENSHOT TO COME

The Logs tab, with Problems in the apps above the log

Alerts

tapiOca alerts Admins, in a pill in the page's header and in the apps, when:

- Plex can't be reached, won't accept the token, or refuses tapiOca's pictures;
- the data folder has less than 2 GB free, or can't be written to;
- a Station keeps failing to start (3 times in 10 minutes), or backups fail twice in a row;
- tapiOca's clock is more than 2 minutes off from Plex's;
- the apps can't reach tapiOca from outside;
- the **Broken files** tab needs an Admin (at most once an hour), or clip folders need ratings;
- an Admin's PIN was entered wrong 5 times; or
- a door is left open (see Home, away and your network).

An alert starts only once a problem persists, and clears once things stay right, so alerts don't flicker. Each start and fix is logged. Alerts survive restarts, and a fix while tapiOca was stopped is still reported, once.

Alert notifications

To get alerts on your phone or in a chat:

1. On the **Logs** tab, open **Alert notifications**.
2. Tick **Send each alert to the addresses below when it starts and when it's fixed**.
3. Enter a **Web address**, and choose its format: **ntfy or plain text** (the default), **JSON (Gotify, Home Assistant and others)**, **Discord** or **Slack** (Mattermost and Rocket.Chat take Slack's too).
4. **Add an address** for more, up to 3.
5. Choose **Send a test**, then **Save**.

JSON sends `{"title": "tapiOca", "message": "...", "kind": "...", "state": "started"}` (or `"fixed"`). A Discord or Slack webhook always gets its own format, and never pings anyone. One address failing doesn't hold up the others. tapiOca waits 5 seconds, tries once more, and follows no redirects. Only Admins see the addresses.

> **Important:** An alert can name a person, a title and a Station, and give your Plex server's home address and your away-from-home address. Send alerts only to services you trust.

Backups, updating and rolling back

Backups

tapiOca backs itself up about 15 minutes after it first starts, then every night at about 3 AM, and keeps the newest 7 in `data/backups`. A backup holds your Stations and schedules, every setting, the Broken files list, the tuner's identity (so Plex still knows it), your logos, people and their passwords (as hashes), and viewing stats. It leaves out Intro Bumper videos, which stay in `data/bumpers`. Your Plex token is in tapiOca's app settings, not its data folder.

On the **Setup** tab, under **Backups**:

- **Download a backup** makes one now and downloads it (it also counts toward the 7 kept).
- **Restore a backup…** replaces everything above with a backup's copy.
- **The list** shows each backup, with **Download**.

> **Important:** Keep backup files private. A backup holds Sonarr's and Radarr's API keys, your alert web addresses (a webhook address is a secret) and password hashes. Only sign-ins and devices are left out.

Restoring. Choose **Restore a backup…**, pick the `.zip`, and confirm. tapiOca checks that it's a genuine tapiOca backup, saves what you have as a "before restore" backup, restarts, and puts the backup in place. Anyone watching is disconnected, and everyone signs in again as the people in the backup. A backup with no people turns sign-in off, so tapiOca asks first.

> **Important:** Keep the container's restart setting (`restart: unless-stopped`). A restore restarts tapiOca, and Docker must start it again.

Knowing when a new version is out

tapiOca never updates itself. With **Tell me when a new tapiOca is out** ticked, under **New versions** on the **Setup** tab, it asks GitHub every 4 hours whether a newer version is out, sending only its version (GitHub sees your internet address, as any website does). It's off until an Admin answers, in the setup's last step or here. **Check now** looks right away. tapiOca 1.44.2 and 1.44.3 can't find newer versions (they ask GitHub by the number of a repository that's gone): update them by hand, as below.

While a newer version is out, Admins see a banner: "tapiOca 1.45.0 is available," with the steps for your install, **What's new**, **Back up now**, **All the steps** and **Not now** (which hides it for that version, in that browser). A "tapiOca 1.45.0 is out" pill stays in the header.

Updating

Back up first (**Back up now**, or **Download a backup**), then:

Install	To update
TrueNAS SCALE	In **Apps**, open **stationplay**, and choose **Update** if it's offered. If not, run `sudo docker pull ghcr.io/egadgetboy/tapioca-server:latest` in **System → Shell**, then **Edit** the app and **Save**. (Not confirmed yet.)
Docker Compose, Docker Desktop	In the compose file's folder, run `docker compose up -d --pull always`.
Unraid	On the **Docker** tab, **Check for Updates**, then **apply update** on **stationplay**.
Synology	Over SSH, in tapiOca's folder, run `sudo docker compose up -d --pull always` (on older systems, `sudo docker-compose pull`, then `sudo docker-compose up -d`). (Not confirmed yet.)
docker run	`docker pull ghcr.io/egadgetboy/tapioca-server:latest`, `docker rm -f stationplay`, then your `docker run` command.

Your data folder, and so your Stations, settings, logos, videos and lists, carries over. Existing Stations keep their settings. The page reloads with the new version, and the setup may open once with new questions.

A database copy first. Before an update changes the database, tapiOca copies it to `data/backups/before-<version>-<date>.db`, and keeps the newest 3. If it can't (a full disk, say), it changes nothing and stops: make room, then start it again. 1.44.3 added one table, so updating to it or later from an earlier version makes this copy. 1.44.4 changes nothing in the database.

Staying on one version. Change the end of `image:` from `:latest` to `:1.44` (fixes for 1.44 only) or `:1.44.4` (exactly that). Change it back to `:latest` to follow every release. Images exist from 1.34.0 on.

From before 1.34.0, when tapiOca built itself, switch once: back up, delete the `build:` block from your file, set `image:` to `ghcr.io/egadgetboy/tapioca-server:latest`, add `TAPIOCA_INSTALL` for your install, and start it again. Your data carries over.

Rolling back

1. **Stop tapiOca** (in TrueNAS's **Apps**, with `docker compose stop`, `docker stop stationplay`, or on Unraid's **Docker** tab).
2. **Put back the database copy,** if the update made one. In the data folder:

`sudo cp backups/before-1.44.3-20261101-153000.db stationplay.db sudo rm -f stationplay.db-wal stationplay.db-shm`

(With your copy's name. In PowerShell, use `Copy-Item` and `Remove-Item`.) 3. **Ask for the previous image,** such as `ghcr.io/egadgetboy/tapioca-server:1.44.2`, in `image:` (or Unraid's **Repository**), and start it.

> **Important:** The copy made before an update holds everything, sign-ins included. Putting it back brings back sign-ins ended since the update (Sign outs included), old passwords and PINs, and people removed since. Check the **Access** tab afterward. Going back to 1.44.2 also brings back the Sign out problem 1.44.3 fixed (see People, sign-in and Viewing Levels), and going back to 1.44.2 or 1.44.3 brings back an update check that can't find newer versions.

Everyone stays signed in, and anything changed since the update is lost. **From 1.44.4 to 1.44.3, or from 1.44.3 to 1.44.2,** skip step 2: the older one runs on this database as it is, and nothing is lost. Each release is signed, with `SHA256SUMS` beside it (the README's "Checking a release's signature" has the commands).

Security and the API

The two ports

Keep port 3310 on your home network. Its tuner, guides and streams never ask for a password, because Plex can't sign in to a tuner. If it's reached from the internet anyway, tapiOca treats those connections as from the internet, and alerts Admins.

The public port (`PUBLIC_PORT`, 3311) is for the page and apps from the internet, behind a reverse proxy with HTTPS or a tunnel. There:

- **No Plex addresses:** the tuner, guides, streams and playlist don't exist.
- **Sign-in is always required.** Visitors who haven't signed in see only the sign-in page, tapiOca's name and version, its notices and privacy notice, and what the apps need to connect. Until signing in is on, it says only "This tapiOca client can't reach the server. Please check back later."
- **Wrong passwords are limited:** 5 in 15 minutes for each address, and 100 for the whole internet.
- **Apps need HTTPS,** and API tokens are refused unless an Admin allows them.

Built-in protections

- **Plex.** The page never sees your Plex token, and can't send its own requests to Plex. tapiOca plays only Plex's own file addresses.
- **Uploads.** Logos must be images and Intro Bumpers videos, each made into a clean copy. Limits: 500 logos, 50 videos, and 2 GB left free on the data folder's disk.
- **The page** runs only its own scripts, can't be embedded in another site, and refuses changes from other sites (each change carries a header only tapiOca's page sends).
- **Backups** restore only if they're genuine tapiOca backups.
- **Outside connections** go only to addresses you typed and, if you said yes, GitHub. Other addresses must be on the internet. None follows a redirect.
- **Sign-ins.** Passwords and PINs are kept as salted hashes, API tokens as fingerprints. An app's stream addresses end with its sign-in.
- **Tested every release.** Each release passes tapiOca's security tests before it's published: every address, from every kind of visitor, and its sign-ins, outside connections and file paths.

If your server is reachable from the internet

- **Update promptly.** Only the newest release gets fixes.
- **Keep Plex up to date** too.
- **At your reverse proxy,** remove any `X-Forwarded-For`, `X-Real-IP` or `Forwarded` a visitor sends, limit sign-in attempts if it can, and never cache signed-in answers (tapiOca marks them `no-store`). List it under **Trusted proxies**.

- **Use long passwords** for Admins.
- **Never forward** your NAS's admin ports, or 3310.

Signs of a break-in: sign-ins, Admins, addresses, devices, apps or API tokens you don't recognize on the **Logs** and **Access** tabs; unexpected files in the data folder (tapiOca never writes programs there); processes in the container other than tapiOca, `ffmpeg` and `ffprobe`; or high processor use while nothing plays. If you find any, stop tapiOca, restore a backup from before, change every Admin password, remove every linked device and API token, change your Plex password, and update.

Reporting a security problem. Report it privately, never in a public issue: on tapiOca's GitHub page (`github.com/egadgetboy/tapioca-server`), choose **Security**, then **Report a vulnerability**. You'll get a reply within 7 days.

tapiOca's API

Optional. Your own scripts, home automation and other players can use tapiOca's API, under `/api/v1`: the server, its Stations with what's on now and next, the guide, each Station's HLS stream, and how tapiOca is doing. An Admin token can also update a Station, check its files, refresh Plex's guide and make a backup. Version 1 only grows: a breaking change would come as a new version, with version 1 kept at least 6 months after. The reference is `docs/api.md` in the repository, with an OpenAPI spec at `/api/v1/openapi.json`. The apps' own addresses, under `/api/internal`, aren't part of it.

API tokens. With sign-in off, the API needs no token at home. Otherwise:

1. On the **Access** tab, open **API tokens**.
2. Enter **What it's for**, such as "Home Assistant", choose a **Scope** (**Viewer (reads only)** or **Admin**) and when it **Expires** (**In 30 days**, **In 90 days**, **In a year** or **Never**; 90 days to start).
3. Choose **Make token**, then **Copy**. It's shown once.

Scripts send it as `Authorization: Bearer <token>`. A token works only with the API, never does more than its Admin can now, and stops when it expires, is revoked (**Revoke**), or its Admin is removed. **Accept API tokens from the internet** (off to start) lets tokens work through the public port, over HTTPS.

Settings reference

Every setting in tapiOca 1.44.4, grouped as the screen groups them.

The YAML or compose file

Environment variables, set when you install. Restart tapiOca after changing one.

Setting	What it does	Choices	Default
`PLEX_URL`	Plex's address. Leave out for no Plex.	An address	None
`PLEX_TOKEN`	Your Plex token.	Text	None
`TAPIOCA_INSTALL`	How your install updates, for the page's steps.	`truenas`, `compose`, `unraid`, `synology`, `docker-desktop`, `docker-run`	None
`TZ`	Time zone for the clock, specials, checks, backups, stats and logs.	Such as `America/Chicago`	`UTC`
`MEDIA_DIR`	Where your media is inside the container.	A folder	`/media`
`PATH_MAPPINGS`	Plex's paths, for unusual layouts.	`plex/path:local/path;…`	None
`HW_ACCEL`	The encoder.	`auto`, `nvidia`, `intel`, `amd`, `cpu`	`auto`
`HW_DEVICE`	Which GPU, with several.	A render node, or an NVIDIA number	Automatic
`PORT`	The home port.	A port	`3310`
`PUBLIC_PORT`	The public port. Must differ from `PORT`.	A port	Off
`AUDIO_LANGUAGE`	Stations' sound and subtitle language.	A language code	`eng`
`FRIENDLY_NAME`	The tuner's name in Plex.	Text	`tapiOca`
`BASE_URL`	Only if Plex reaches tapiOca at another address than your browser.	An address	Automatic

Setting	What it does	Choices	Default
`IDLE_GRACE_SECONDS`	How long a Station runs after its last viewer leaves.	Seconds	`20`
`AUDIO_BITRATE_KBPS`	Stations' sound bitrate.	kbps	`192`
`DATA_DIR`	The data folder inside the container. Leave it.	A folder	`/data`
`FFMPEG_PATH`, `FFPROBE_PATH`	A custom ffmpeg build.	A program	`ffmpeg`, `ffprobe`

`TUNER_COUNT` and `VIDEO_HEIGHT`, from before 1.14, are read once if set, then ignored.

Setup tab

Setting	What it does	Choices	Default
Picture for new stations	New Stations' picture size.	**480p**, **720p**, **1080p**	**720p**
Tuners	Stations that can play at once.	1 to 20	4
A Media folder's switch	Whether the apps show it.	On, off	On
Name (Add a Media folder)	Its name.	Up to 60 characters	Empty
It holds	What it holds.	**Shows**, **Movies**	**Shows**
Its folders	The folders it reads.	One or more, none inside another	None
Tell me when a new tapiOca is out	The update check.	On, off	Off until answered

The Station editor

The setup's **New station settings** step sets what new Stations start with for **Subtitles**, **Commercials & trailers**, **Station ID card**, **Intro Bumper** (its **Length**), **Up Next Banner**, **In the corner** and **Corner**. The defaults below apply until you change them there.

Setting	What it does	Choices	Default
Number	Its number (99 is INFO's).	1 to 9999	The next free one
Name	Its name.	Up to 60 characters	"Station" and its number

Setting	What it does	Choices	Default
Logo	Its logo.	Built-in, your own, Plex's, **Station number**	Its number's (1 to 50), or an unused one
What's on this station	What it plays.	**Pick shows & movies**, **Use a filter**	**Pick shows & movies**
Picture (How it plays)	Its picture size.	**480p**, **720p**, **1080p**	The new-Station size
Order	How programs follow.	**Episode order**, **Shuffle**	**Shuffle**
4:3 shows	How 4:3 fills the screen.	**Black bars**, **Stretch**, **Zoom**	**Black bars**
Intros & credits	Skip Plex's markers.	**Play them**, **Skip them**	**Play them**
Marathons (Specials)	3 episodes in a row.	**Off**, **Random times** (1 to 4 a week), **Set times**	**Off** (2 a week; Saturday, 8:00 PM)
Which episodes	Where marathons start.	**Next in order**, **Random starting point**	**Next in order**
Feature Presentation	A movie night.	**Off**, **On**, with days and a time	**Off** (Friday, 8:00 PM)
Movies from	Its movies.	This Station's, a Media folder, a collection	This Station's movies
Blocks	Time-of-day blocks.	Up to 4: name (40 characters), days, 30 minutes to 12 hours, shows	None
Commercials & trailers (Between programs)	Clips after each program.	**None**, **1**, **2**, **3**	**None**
Station ID card	A card after each program.	**Off**, **3 sec**, **5 sec**, **10 sec**	**Off**
Jingle	The card's jingle.	**On**, **Off**	**On**
In the corner (On screen)	During programs.	**Nothing**, **Logo**, **Name**, **Clock**	**Logo**
Size	Its size.	**Small**, **Medium**, **Large**	**Large**
Transparency	How see-through.	**High**, **Medium**, **Low**	**Medium**
Corner	Which corner.	**Top left**, **Top right**, **Bottom left**, **Bottom right**	**Bottom left**
Clock	The clock's format.	**12-hour**, **24-hour**	**12-hour**

Setting	What it does	Choices	Default
Look	The logo's color.	**In color**, **All white**	**In color**
When	When it shows.	**Always**, **At the start**	**Always**
Up Next Banner	How long it shows.	**Off**, **3 sec**, **5 sec**, **10 sec**	**10 sec**
Size (banner)	Its size.	**Small**, **Medium**, **Large**	**Large**
Subtitles	Drawn into the picture.	**Off**, **Forced only**, **Always**	**Off**
Viewers join (When someone tunes in)	Where they join.	**In progress**, **From the beginning**	**In progress**
Intro Bumper	What plays first.	**Off**, **Built-in card**, **Your video**	**Built-in card**
Length	The card's length.	**3 sec**, **5 sec**, **10 sec**, **15 sec**	**10 sec**
Sound	The card's dial sound.	**On**, **Off**	**On**
Description	The card's line.	Up to 140 characters	Empty
Video	Your video.	Up to 30 seconds, 500 MB	None

Access tab

Setting	What it does	Choices	Default
Name (Add a user)	Their name.	Up to 40 characters, unique	None
Password	Their password.	At least 8 characters (optional for a User)	None
PIN (optional)	Asked when they pick themselves.	4 digits	None
Role	What they can do.	**User**, **Admin**	**User** (the first is an Admin)
Viewing Level	What they can see.	Any level	**New people start on**'s
Stations they can make	A User's Stations.	**None (watches only)**, 1, 3, 5, 10, 25, **Any number**	3

Setting	What it does	Choices	Default
Can change their own password, **Can change their own PIN**, **Can report problems**	A User's permissions.	On, off	On
How you sign in (Your account)	An Admin's sign-in.	**Full**, **Light**	**Full**
Trusted networks	Skip signing in.	Up to 50, giving **Who's tuning in?** or **Signed in as** a person	None
Trusted proxies	Believed about visitors.	Up to 10 home addresses	None
Names for tapiOca at home	Home names.	Up to 10	None
New people start on	New Users' level.	Any level	**Unrestricted**
Movies up to, **TV up to**	A level's ratings.	A rating, **No limit**	As each level
Unrated titles	A level's unrated titles.	**Shown**, **Hidden**	**Hidden** (but **Unrestricted**)
Media folders (a level)	What a level sees.	**All**, or some	**All**
Genres, **Tags**, **Collections**, **Studios**	Names a level hides.	Any names	None
Commercials and trailers	Clip folders' ratings.	TV or movie ratings	**Not rated**
Tuner access	Who uses the tuner.	**Any device on my home network**, **Only these devices** (up to 50)	**Any device on my home network**
Stop Plex users from watching stations blocked for them	Plex blocking.	On, off	Off
Let tapiOca's apps watch away from home	The apps away.	On, off	Off
The address apps reach tapiOca at from outside	Their address.	Up to 200 characters	Empty
Media away from home	Media's quality away.	**Original**, **Up to…** 1 to 200 Mbps	**Original**
Devices watching at once	The apps' limit.	A number, or empty	No limit
Away from home, at most	The apps' limit away.	A number, or empty	No limit

Setting	What it does	Choices	Default
Media folders (Media in tapiOca's apps)	Shown in the apps.	Tick each	Every one
When playing can't keep up, At home:	Media stalling at home.	**Stop, and offer a smaller version**, **Switch to a smaller version on its own**	**Stop, and offer a smaller version**
When playing can't keep up, Away from home:	Media stalling away.	The same	**Switch to a smaller version on its own**
Even sound for a show's episodes	Even loudness in Media.	On, off	On
Accept API tokens from the internet	Tokens on the public port.	On, off	Off

Logs, Broken files and the footer

Setting	What it does	Choices	Default
Send each alert to the addresses below when it starts and when it's fixed	Alert notifications.	On, off, up to 3 **Web address** entries	Off
Format	Each address's format.	**ntfy or plain text**, **JSON (Gotify, Home Assistant and others)**, **Discord**, **Slack**	**ntfy or plain text**
Deep scan every night from	The overnight scan.	On, off, with times	On, 1:00 AM to 6:00 AM
Use Sonarr for episodes, Use Radarr for movies	File replacing.	On, off, with an address and **API key**	Off
Files to replace	Which entries.	**Broken or damaged**, **Missing**, **Both**	**Both**
When to replace	Ask first or not.	**Only when I ask**, **Automatically**	**Only when I ask**
Appearance (footer)	Light or dark, in one browser.	**Automatic**, **Light**, **Dark**	**Automatic**

The data folder

`stationplay.db` (the database), `broken-files.json` (the Broken files list), `device.json` (the tuner's ID in Plex), `cuts.json` (where skips end), `backups/`, `logos/` and `bumpers/`. An empty `reset-access` file, then a restart, removes every person.

Troubleshooting

Start with the **Logs** tab: search it, or tick **Warnings** or **Errors**, and **Copy** what it says. `docker logs stationplay` has more, back to Docker's last rotation (3 files of 10 MB each, as installed). A good bug report, as an issue on GitHub, says what you expected, what happened, and what the log showed.

Installing and starting

Symptom	Cause	Fix
It keeps restarting, and can't write to `/data`.	The data folder isn't owned by tapiOca's user.	Run the `chown` command the message shows, with your folder, then restart.
It won't start, and mentions `/dev/dri`.	GPU lines, but no Intel or AMD graphics.	Remove the `devices:` and `group_add:` lines.
Port 3310 or 3311 "is already allocated".	Another app uses it.	Change the port line's first number (`"3320:3310"`), and use it everywhere.
The page still shows the old version after updating.	Docker didn't fetch or use the new image.	Check that `image:` ends in `:latest`, then pull it yourself and start again (see Updating).
"No matching manifest" for your system.	The image is 64-bit only.	Use an x86-64 or ARM64 machine, and a 64-bit OS on a Pi.
Nothing to play yet, or "Plex didn't accept the token in PLEX_TOKEN".	No Media folder, or Plex isn't set up.	Add a Media folder, or fix `PLEX_URL` and `PLEX_TOKEN`. Use Plex's network IP, not `localhost`.
The first check finds nothing at `MEDIA_DIR`.	The media folder isn't mounted there.	Fix the media `volumes` line, or set `MEDIA_DIR` as the check says, restart, and **Check again**.

Plex

Symptom	Cause	Fix
Plex isn't connected.	Its address, Plex itself, or the token.	Check `PLEX_URL` (not the example), that Plex runs, and `PLEX_TOKEN`. The setup's **Tuner** step says which.
Plex can't find the tuner.	tapiOca doesn't answer Plex's search.	Enter the tuner address yourself.

Symptom	Cause	Fix
Plex says no tuners are available.	Plex hasn't read the new number.	Restart Plex after raising **Tuners**.
The guide shows the wrong program.	Plex has an older guide.	**Refresh Guide** in Plex's tuner settings, wait a minute, and reload.
"Device not found".	Plex knows the tuner by an old ID.	Compare the log's tuner ID with `DeviceID` at `http://<server-ip>:3310/discover.json`. Put Plex's ID in `data/device.json` as `{"deviceId": "1A2B3C4D"}`, owned by tapiOca's user, and restart.
A managed user can't see the Stations.	Their Live TV access is too narrow.	Set it to **Allow Live TV and DVR access**.
Skipping intros skips nothing.	Plex hasn't marked any yet.	Turn on Plex's marker settings (see How Stations play).

Playing

Symptom	Cause	Fix
"This station is off the air. Please report it."	Nothing on it could play, or its stream kept failing.	The log says why: usually Plex or the share is down, or every program is broken.
A stream stops, or "is falling behind real time".	The server can't convert fast enough, or the viewer left.	Use a smaller picture, fewer tuners, or a GPU. Run **Test this server**.
HDR looks washed out.	ffmpeg can't convert HDR.	Use tapiOca's own image.
Subtitles don't show.	None in `AUDIO_LANGUAGE`, or not forced ones.	Add subtitles in that language. The first airing may play without them.
No commercials or trailers.	Folders not found, or not rated.	Name them exactly `commercials` and `trailers`, at the top of a Media folder, inside `/media`. Rate them for people with limits.
A special doesn't happen.	A more important one overlapped, or it was too late.	The card shows when the next is, and the log says why.
Up Next Banners never show, and the log names the data folder.	`DATA_DIR` has characters ffmpeg can't handle.	Remove `DATA_DIR`.
A Station stopped following Plex.	An update removing over half its programs waits.	Choose **Update now** if it's right.

Files, GPU and access

Symptom	Cause	Fix
Many files listed at once.	Plex lost them, or a share was down.	Check the **Media files** line, fix the cause, then **Retry**.
Damaged, but it plays fine for you.	Something you can't see.	**Retry**: the checks then leave it alone.
A file or folder is skipped in a Media folder.	Its name isn't UTF-8.	Rename it, then **Read again**.
The GPU isn't used.	The group number, or `/dev/dri` not passed in.	The **Video encoding** line says why. `docker exec stationplay vainfo` shows what the driver sees.
The GPU "was turned off after 3 programs in a row failed on it".	tapiOca set it aside.	Restart tapiOca. If it keeps happening, check the driver.
Locked out of the page.	No Admin can sign in.	Use `reset-access` (see People and sign-in).
A trusted device still asks to sign in.	A non-home name, a proxy, or Docker hiding addresses.	The **Access** tab says why. Open tapiOca by a home address or name.
Away from home is **Down**.	The apps can't reach tapiOca.	Read the status. Point the proxy at `3311` and list it under **Trusted proxies**.
Away from home is **Can't check**.	Your router doesn't loop back.	Test from mobile data, or add a DNS host override.
"Stopping playback needs Plex Pass..." or "Plex won't say who's watching...".	No Plex Pass, or a token that can't see sessions.	Use Plex Pass, and the server owner's token.

Glossary

Admin
A person who can do everything in tapiOca. There's always at least one.

Alert
A note to Admins that something needs attention, shown in the header and log, and sent to up to 3 addresses if you choose.

API token
A secret, shown once, that lets a script use tapiOca's API.

Broken files list
The list of files that can't play properly. They're kept off the air on every Station.

Burn-in protection
The slight, steady movement of what's in the corner, so it can't mark an OLED or plasma TV.

Converted copy
A copy made as a title plays, with the picture made again, for a device that can't play the file.

Deep scan
The overnight check that decodes each file in full.

Dolby Vision profile 5
Dolby Vision with no standard picture inside. Stations list it as **Unsupported**.

Even sound
Episodes brought to the same loudness, −24 LUFS, so the volume doesn't jump.

Feature Presentation
A special: a movie night at set times.

Full sign-in, Light sign-in
An Admin's choice: a password every time (Full), or a password once on each device, then a PIN there at home (Light).

GPU
A graphics processor that converts video with far less work. tapiOca uses Intel, AMD and NVIDIA GPUs.

Guide
What's on each Station over the next 2 days, given to other apps as XMLTV.

HDR, SDR
High and standard dynamic range: two kinds of picture color. Stations play in SDR.

HLS
A video format many players use. Each Station is available as HLS at home.

Home port, public port
Port 3310, for home only (the page, Plex's tuner and the streams), and port 3311, for the page and apps from the internet, where sign-in is always required.

INFO station
Station 99, which comes with a new tapiOca: what's new, tips and counts.

Intro Bumper
A card or video that plays when a Station's stream starts.

Invite code
A one-time code, good for 7 days, to get someone on a device without a password.

Linked device
A device that opens on **Who's tuning in?**, because someone signed in on it.

M3U playlist
The list of Stations other apps read, at `/stations.m3u`.

Marathon
A special: three episodes of one show in a row.

Media
Your movies and shows, on demand in tapiOca's apps.

Media folder
One collection of your shows or movies: Plex's, or one tapiOca reads itself.

NAT loopback
A router feature that lets home devices use the home's internet address. Without it, tapiOca can't check its outside address from home.

NFO file
A small file beside a video with its title, plot, rating and other details.

PIN
A 4-digit number a person gives when they pick themselves on a shared device.

Plex Pass
Plex's paid plan, needed to watch Stations in Plex, find intro markers, and block Stations for Plex users.

Quick check
A fast check that a file opens, and its picture and sound decode at five points.

Repackaged copy
A copy that keeps the picture as it is and converts only the sound.

Restricted program
A program someone may not watch, blocked for them while it airs.

Reverse proxy
A program that takes HTTPS from the internet and passes it to tapiOca's public port.

Smart stations
Stations made in one step from a filter, one or one for each value.

Stand-in
A program that plays for the rest of a failed program's time.

Station
A live, always-on stream of your own movies and shows, with a number, a name, a logo and a guide.

Station ID card
A short card after each program, with a jingle.

Trusted network, trusted proxy
A network whose devices skip signing in; a proxy whose word about its visitors tapiOca believes.

Tuner
What other apps use to watch your Stations; also each Station tapiOca can play at once.

Up Next Banner
A banner near each program's end saying what's next.

User
A person who watches what their Viewing Level allows, and can make a few Stations.

Viewing Level
What a person can see, by rating, Media folder, genre, tag, collection or studio.

VPN
A private network, such as Tailscale or WireGuard, that reaches tapiOca as if at home.

Who's tuning in?
The screen a shared device opens on, where each person picks themselves.

XMLTV
The guide format other apps read, at `/xmltv.xml` (Plex) and `/guide.xml`.

Notices

The license

tapiOca is Copyright © 2026 egadgetboy, licensed under the GNU Affero General Public License, version 3 (AGPL-3.0). In short: you may use, study, change and share it, even for profit. If you share it, or run a changed copy others use over a network, offer its source under the same license (and point the page's **Source** link to it). Keep the notices, and mark what you changed. tapiOca comes with no warranty. The `LICENSE` file in the repository is what applies. tapiOca, its logos, jingles, sounds and INFO film are by egadgetboy. Other product names are the property of their owners.

The privacy notice

The full notice is served by your server at `/privacy` (**Privacy**, at the bottom of the page), and is `PRIVACY.md` in the repository. In short:

- **tapiOca's developer runs no service for it** and receives nothing from it: no accounts, analytics, ads or tracking.
- **It keeps, in its data folder:** your Stations and settings (with Sonarr's and Radarr's keys and alert addresses); people, their Viewing Levels, sign-ins and devices, with passwords and PINs as salted hashes; what each person watches, and their stats; Plex users' names and what they watched on your Stations; the access log; trusted networks, proxies and names; reported problems; and backups. Admins can see all of it. Stats are cleared after 400 days, the apps' problems after 30, and handled reports after 90.
- **Removing a person** removes their sign-ins, Resume places and languages. Their stats and access-log lines stay, apps they linked keep their name until those apps are removed, their reports and their apps' problems stay until cleared, and backups keep everything until replaced.
- **It contacts** only your Plex, your alert addresses, Sonarr and Radarr, your own away-from-home address, and, if an Admin says yes, GitHub every 4 hours, sending only its version.

Plex's own privacy policy covers Plex.

Open-source notices

The full list is at `/open-source-notices` on your server (**Open-source notices**, at the bottom of the page), and in `THIRD_PARTY_NOTICES.md`.

- **FFmpeg,** for all video work, under the GNU GPL, version 2 or later. Each release carries the GPL and LGPL source.
- **Fonts:** DejaVu (Bitstream Vera Fonts license); Montserrat, Space Grotesk, Space Mono and Unbounded (SIL Open Font License 1.1).
- **Common passwords,** from the UK NCSC's list via SecLists (MIT License, Copyright (c) 2018 Daniel Miessler). Drivers, Debian, Python and its packages keep their own licenses.